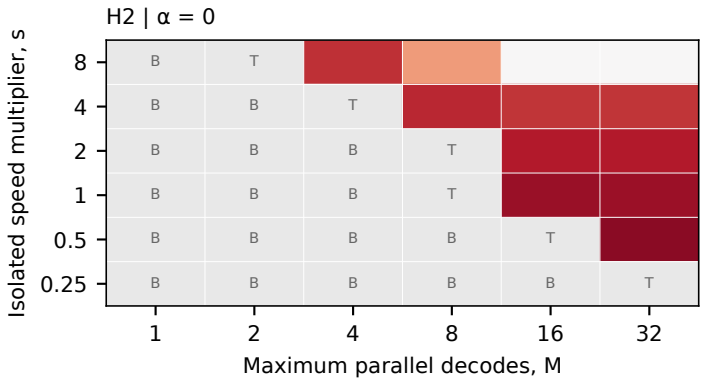
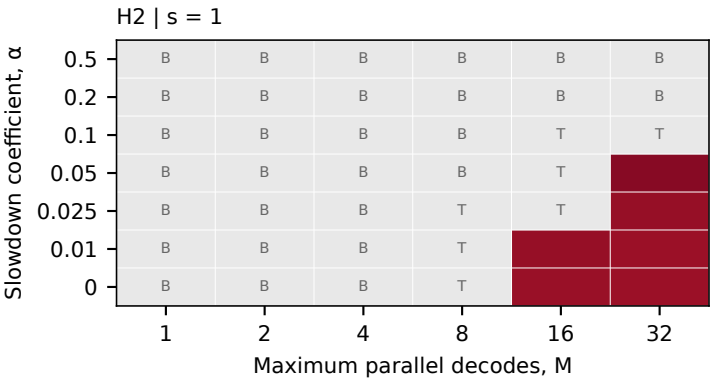
aligned · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

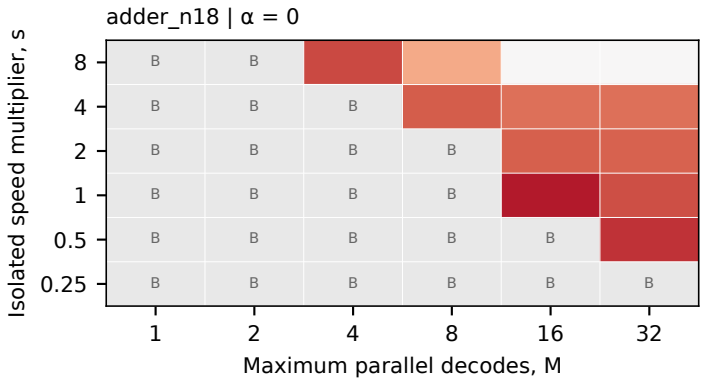
H2 | α = 0



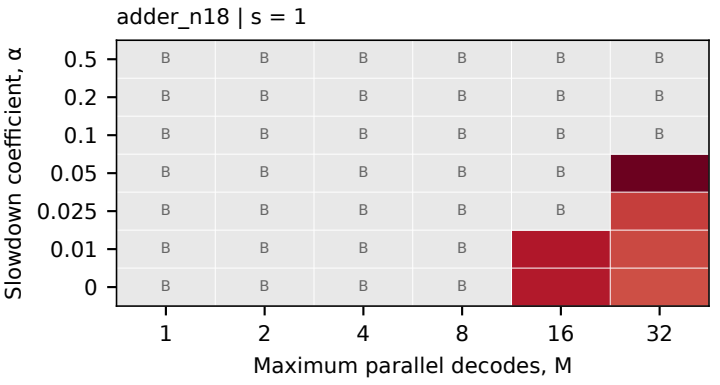
H2 | s = 1



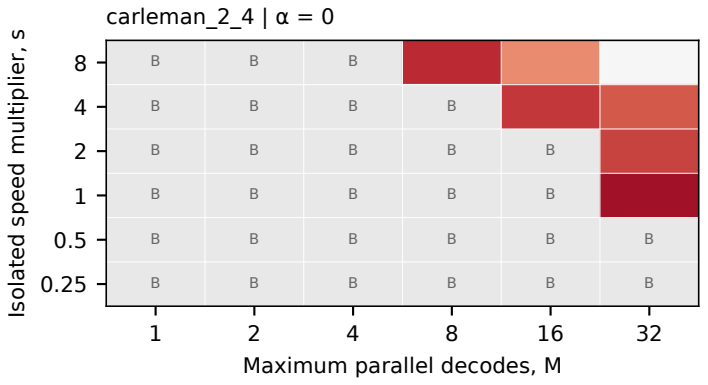
adder_n18 | α = 0



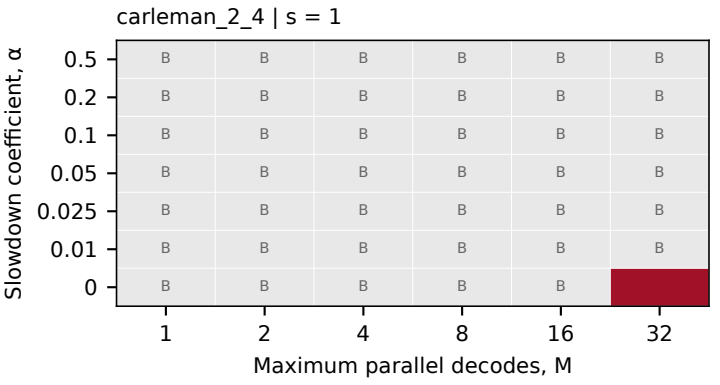
adder_n18 | s = 1



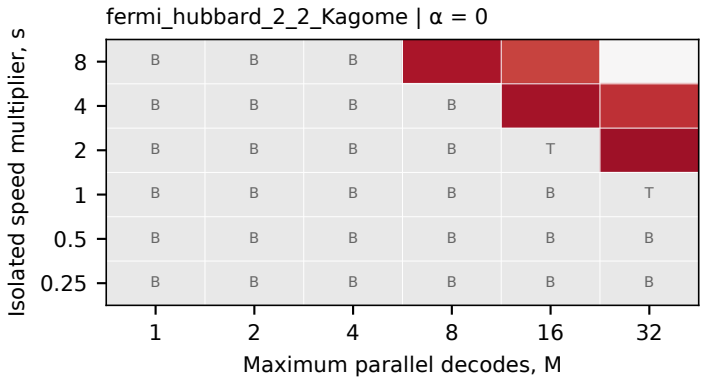
carleman_2_4 | α = 0



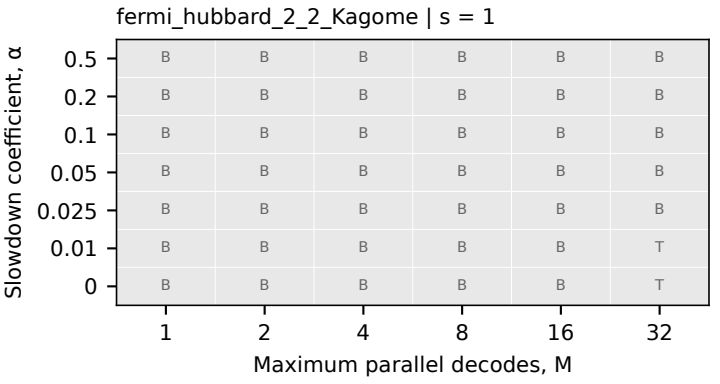
carleman_2_4 | s = 1



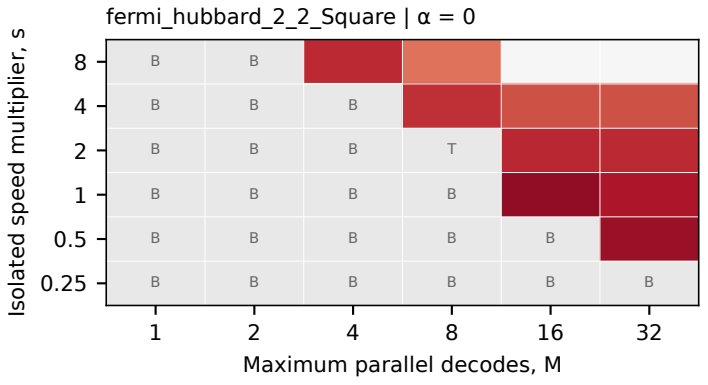
fermi_hubbard_2_2_Kagome | α = 0



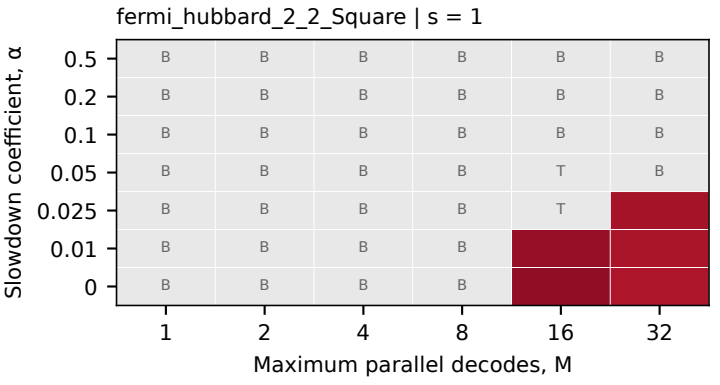
fermi_hubbard_2_2_Kagome | s = 1



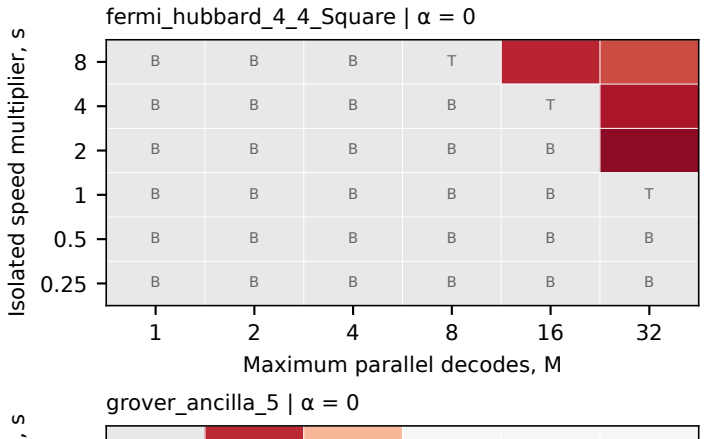
fermi_hubbard_2_2_Square | α = 0



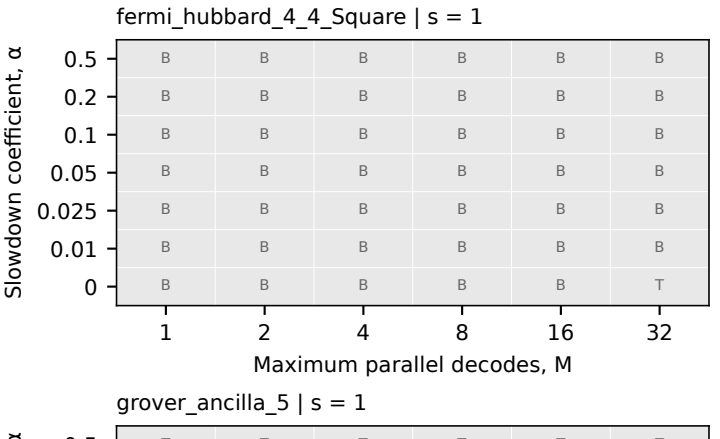
fermi_hubbard_2_2_Square | s = 1



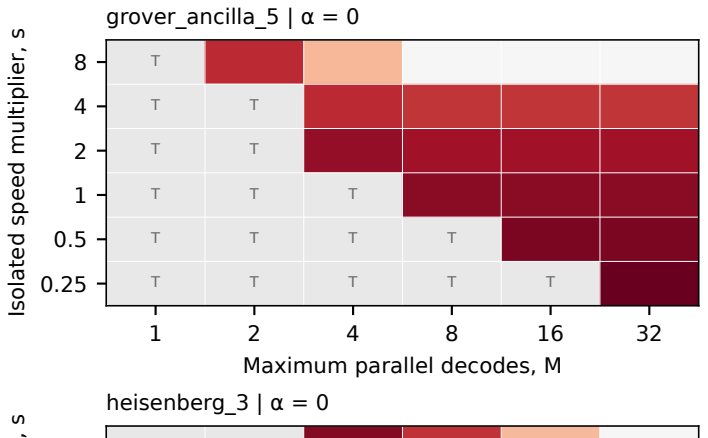
fermi_hubbard_4_4_Square | α = 0



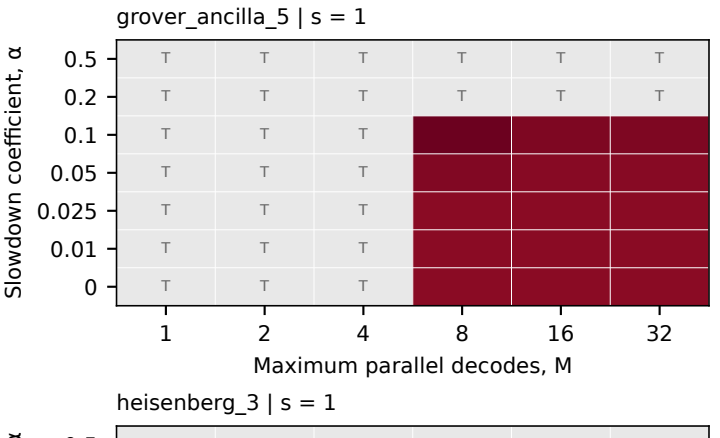
fermi_hubbard_4_4_Square | s = 1



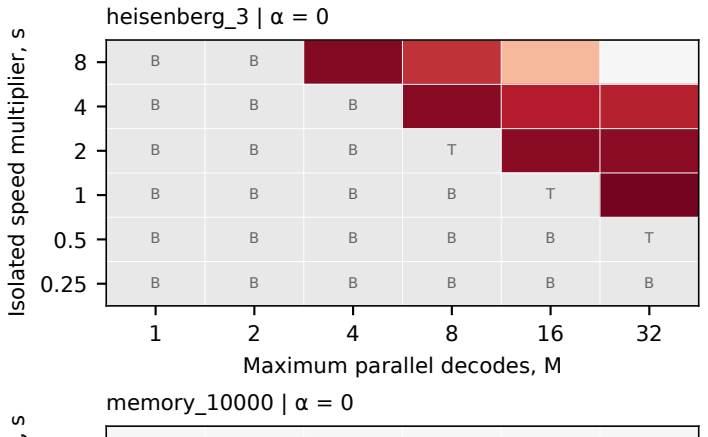
grover_ancilla_5 | α = 0



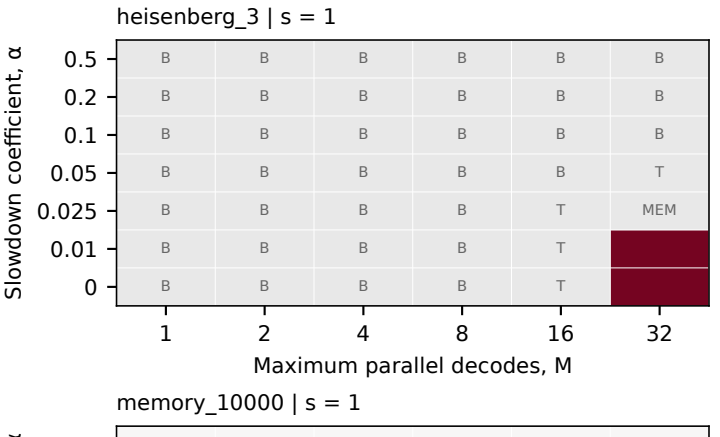
grover_ancilla_5 | s = 1



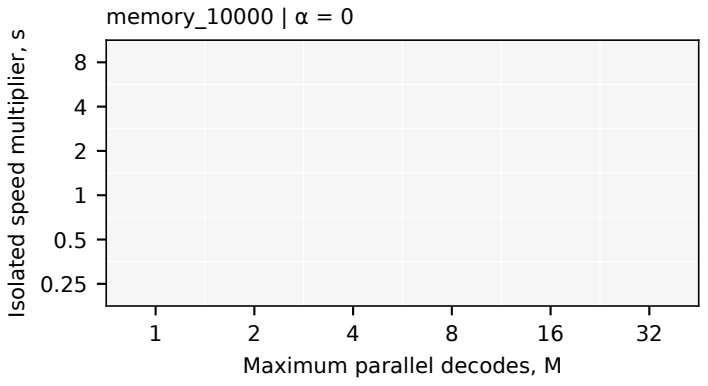
heisenberg_3 | α = 0



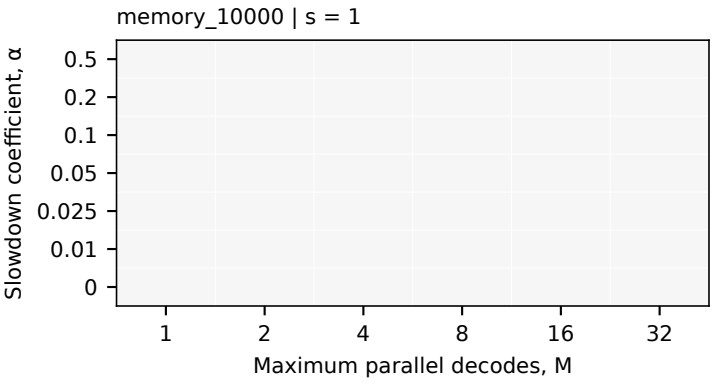
heisenberg_3 | s = 1

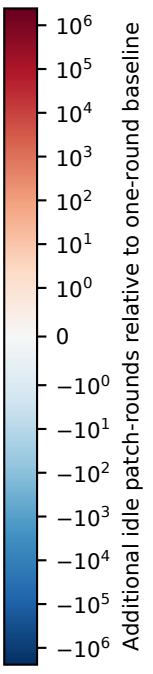


memory_10000 | α = 0



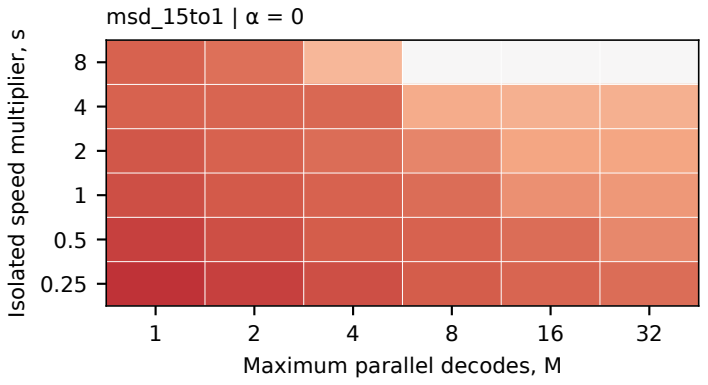
memory_10000 | s = 1



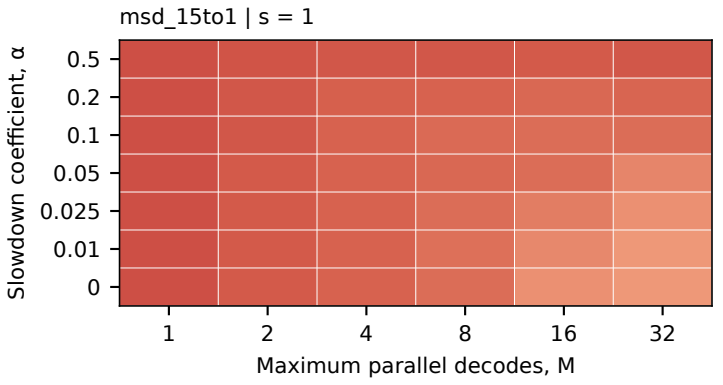


aligned · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

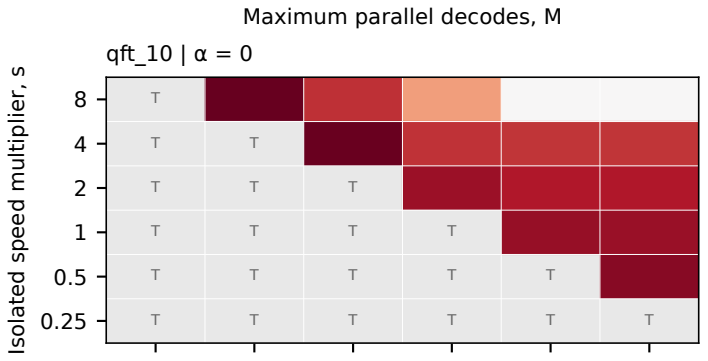
msd_15to1 | α = 0



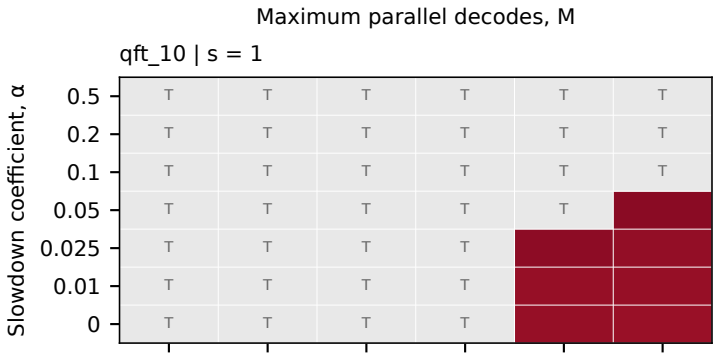
msd_15to1 | s = 1



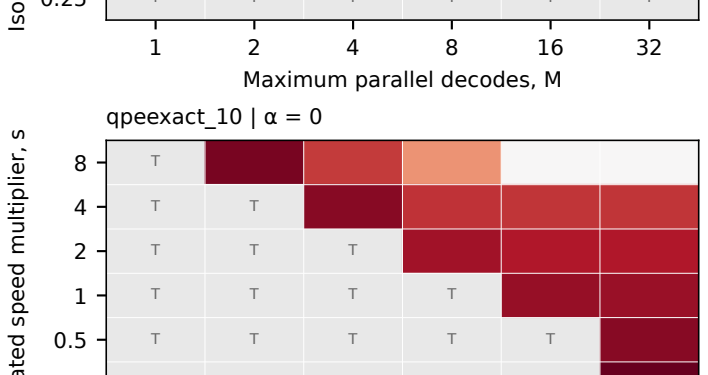
qft_10 | α = 0



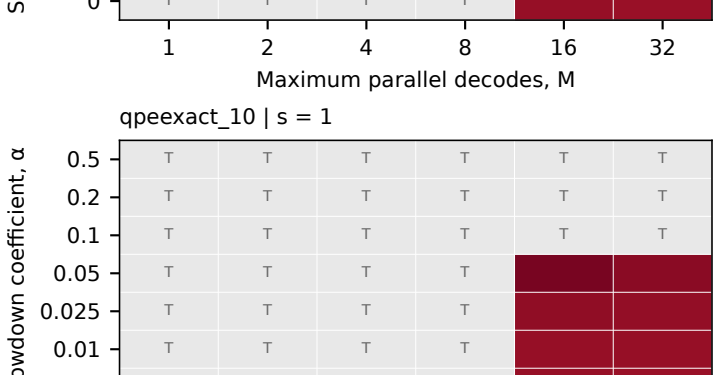
qft_10 | s = 1



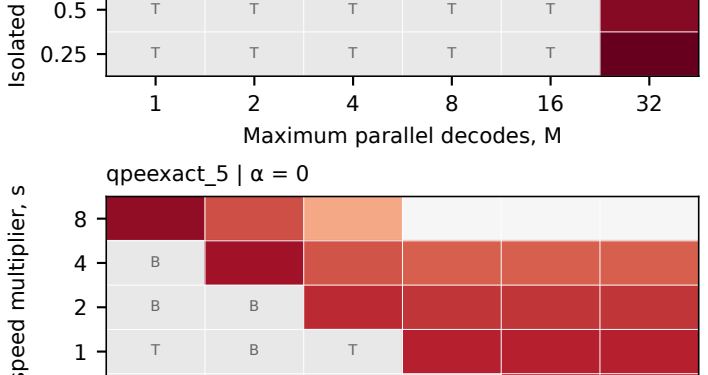
qpeexact_10 | α = 0



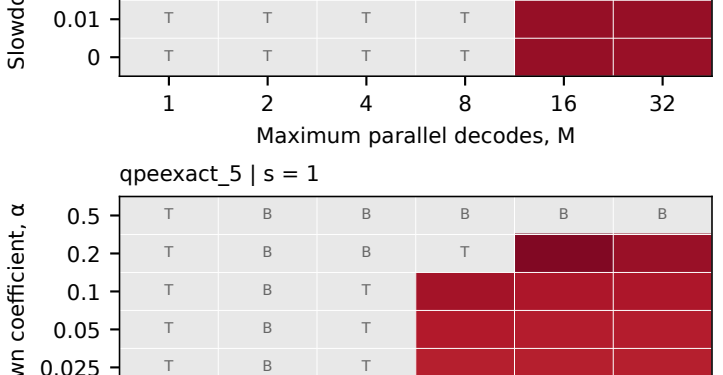
qpeexact_10 | s = 1



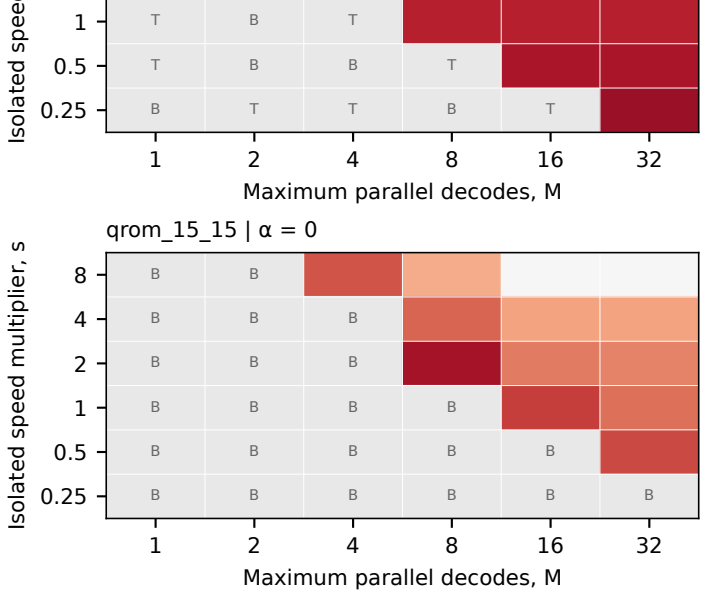
qpeexact_5 | α = 0



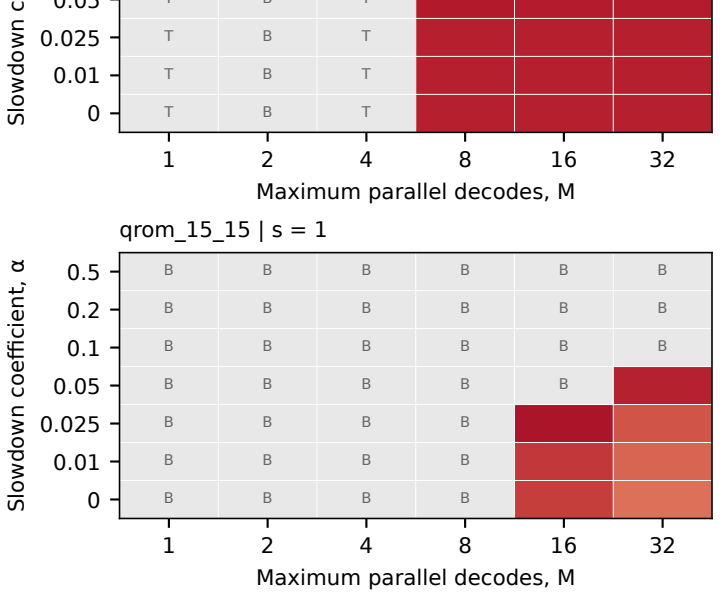
qpeexact_5 | s = 1

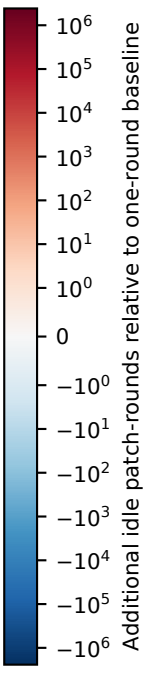


qrom_15_15 | α = 0

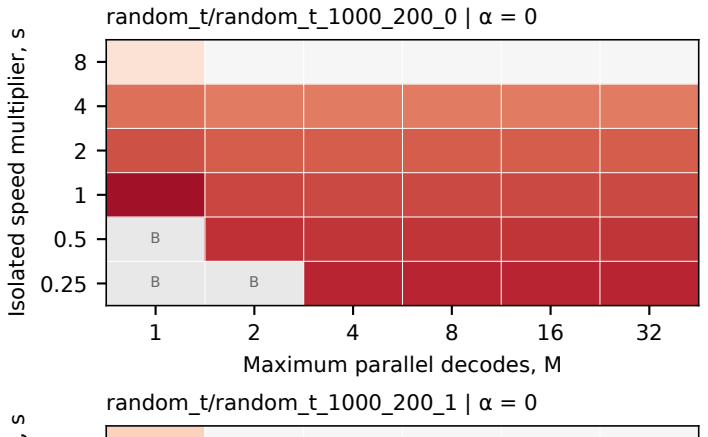


qrom_15_15 | s = 1

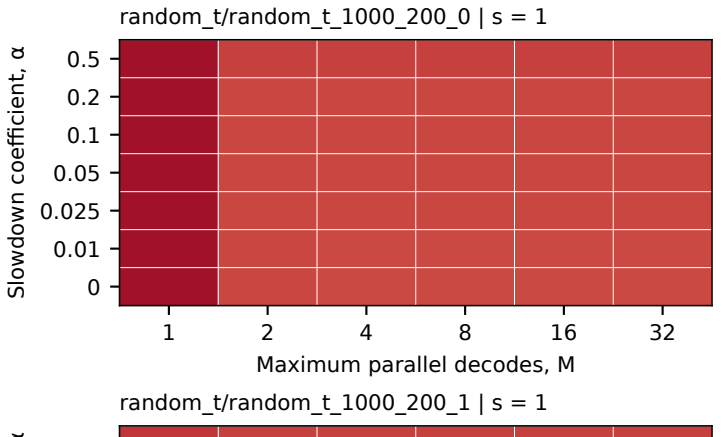




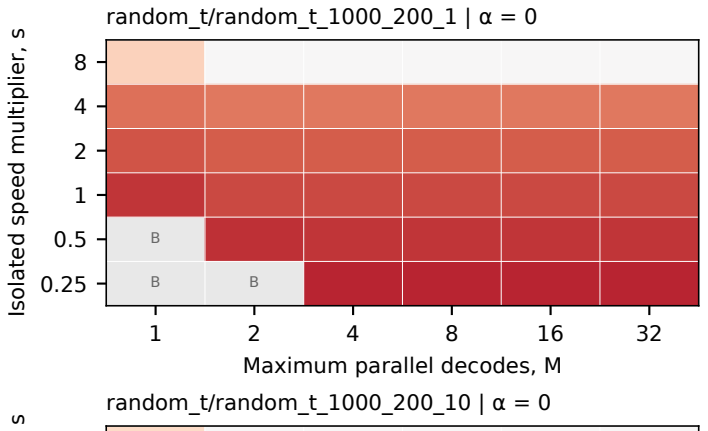
random_t/random_t_1000_200_0 | α = 0



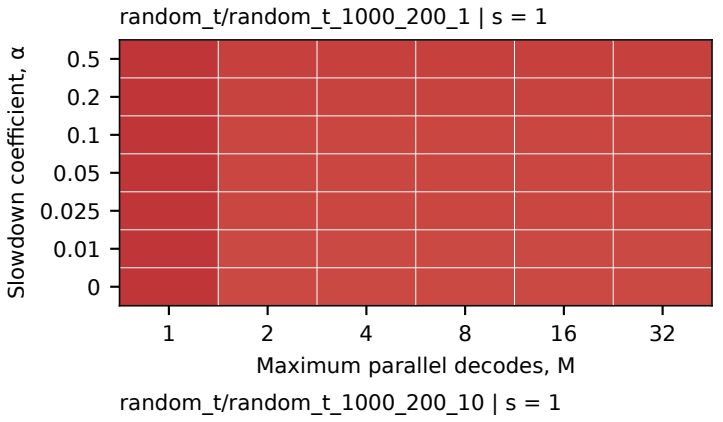
random_t/random_t_1000_200_0 | s = 1



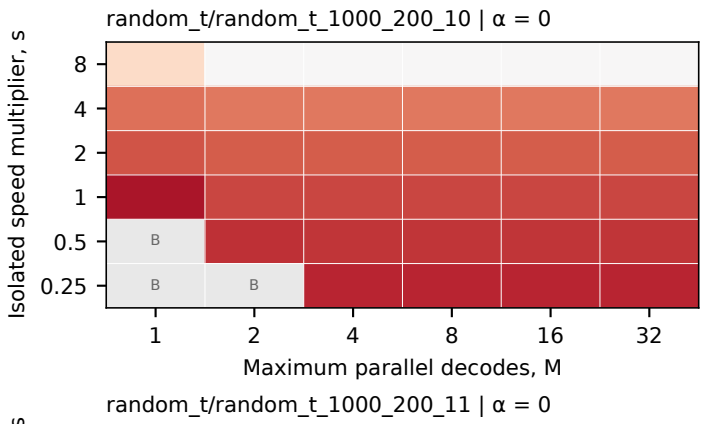
random_t/random_t_1000_200_1 | α = 0



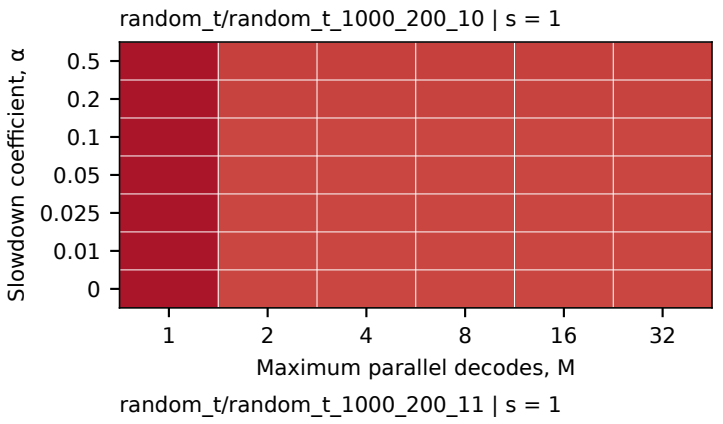
random_t/random_t_1000_200_1 | s = 1



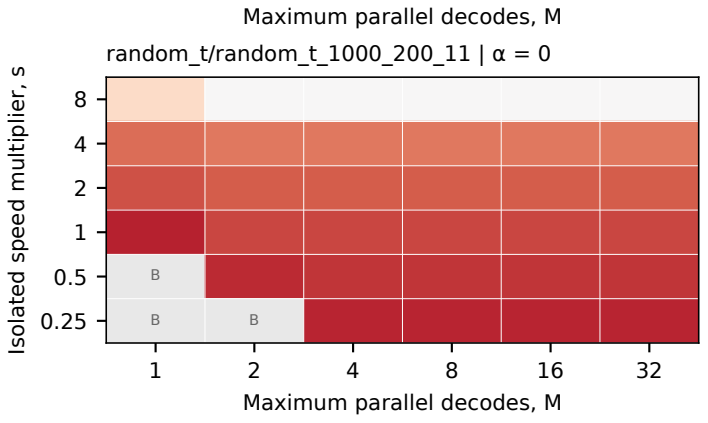
random_t/random_t_1000_200_10 | α = 0



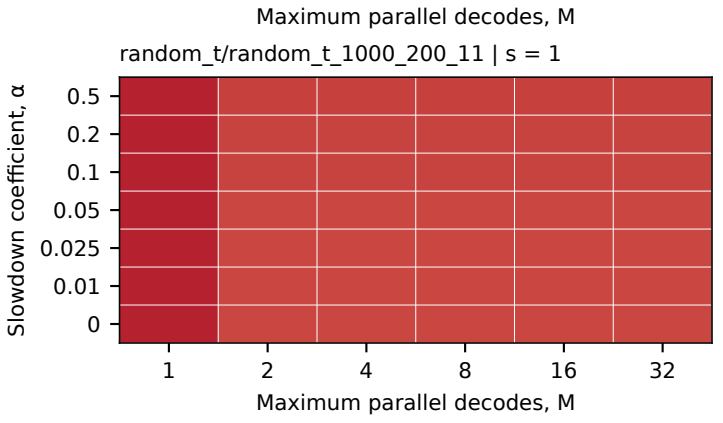
random_t/random_t_1000_200_10 | s = 1



random_t/random_t_1000_200_11 | α = 0

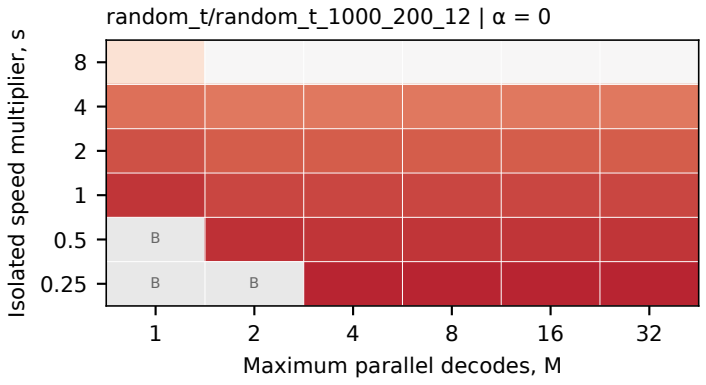


random_t/random_t_1000_200_11 | s = 1

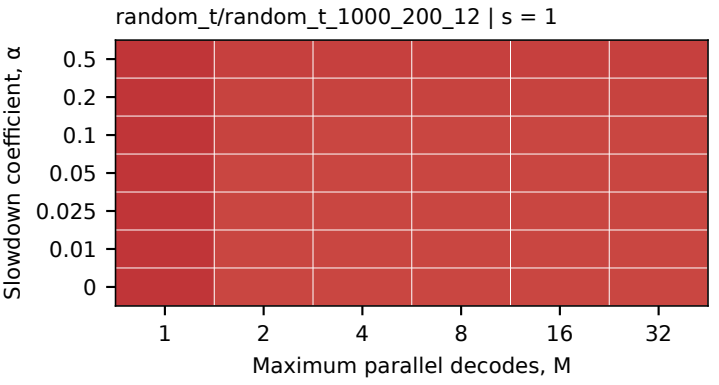


aligned · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

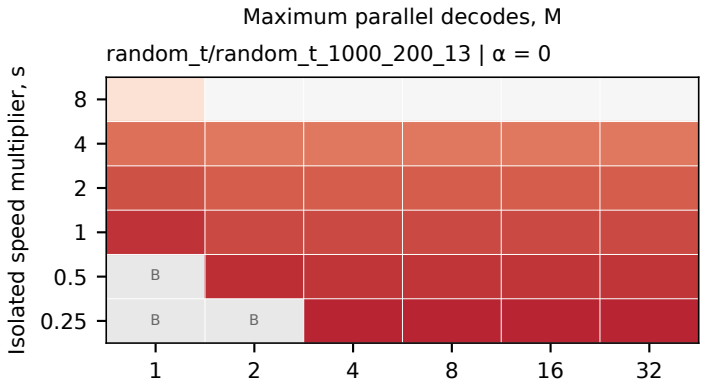
random_t/random_t_1000_200_12 | α = 0



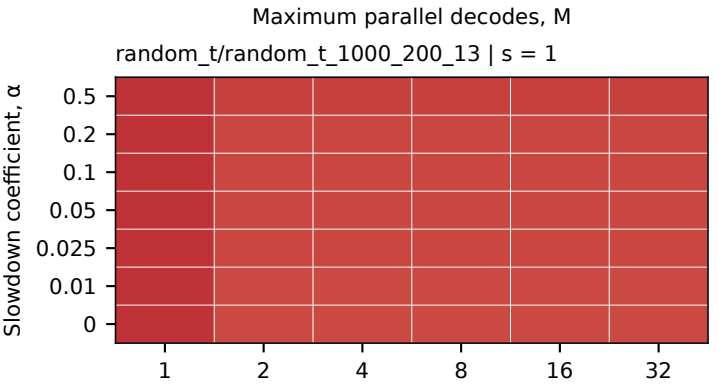
random_t/random_t_1000_200_12 | s = 1



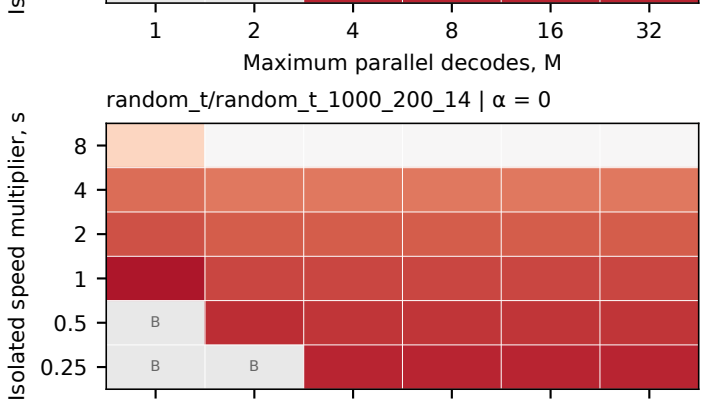
random_t/random_t_1000_200_13 | α = 0



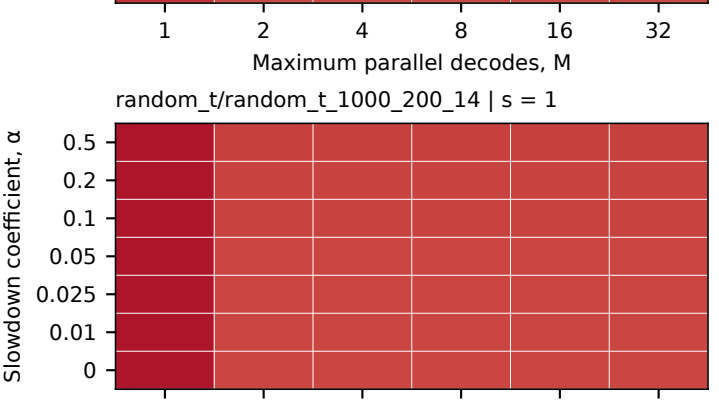
random_t/random_t_1000_200_13 | s = 1



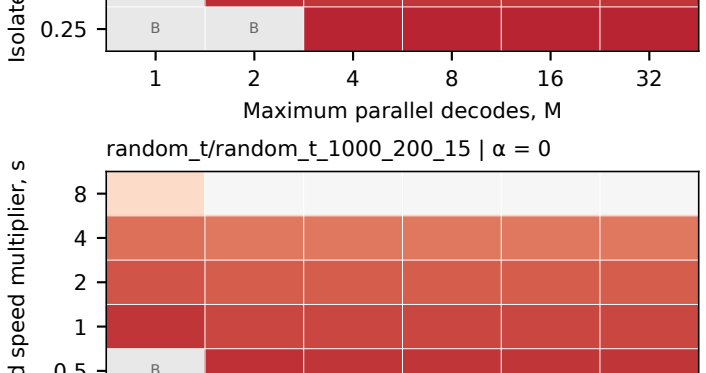
random_t/random_t_1000_200_14 | α = 0



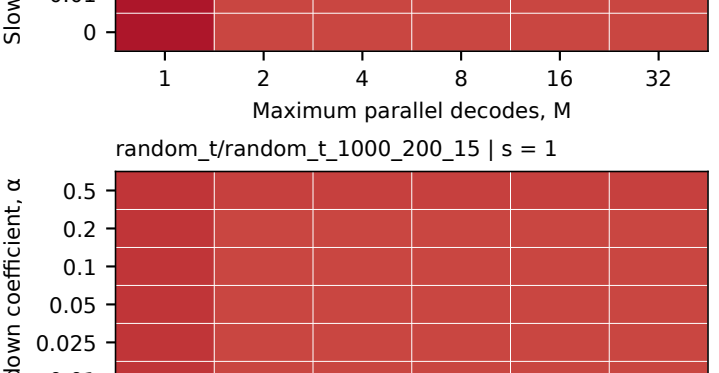
random_t/random_t_1000_200_14 | s = 1



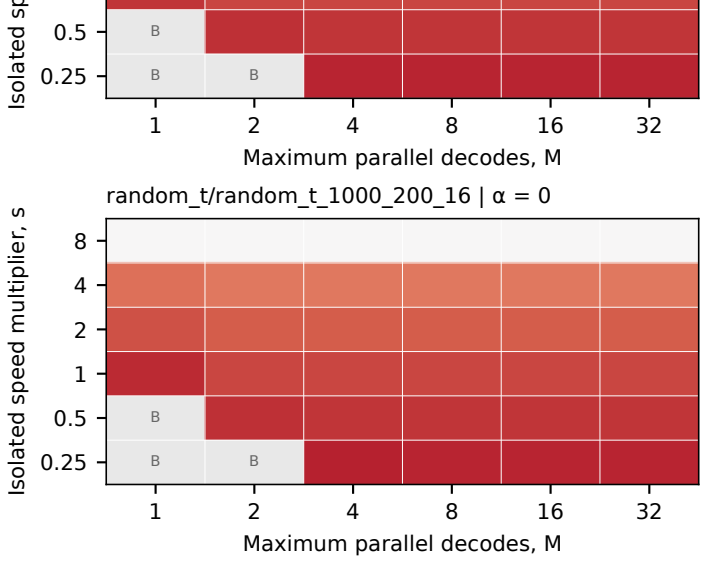
random_t/random_t_1000_200_15 | α = 0



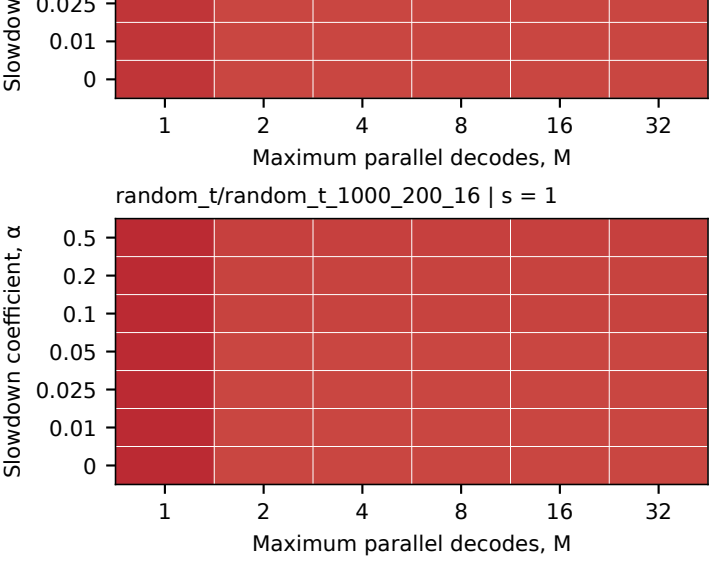
random_t/random_t_1000_200_15 | s = 1

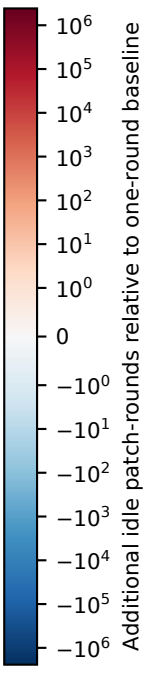


random_t/random_t_1000_200_16 | α = 0

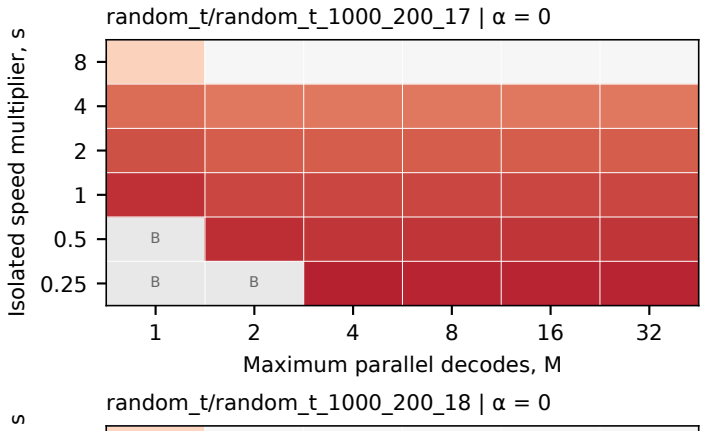


random_t/random_t_1000_200_16 | s = 1

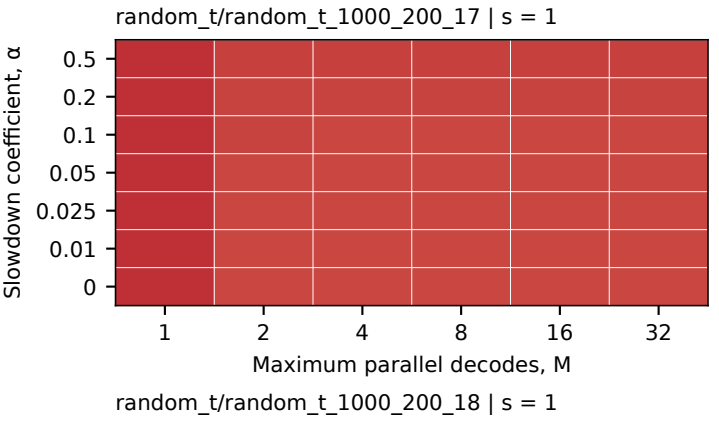




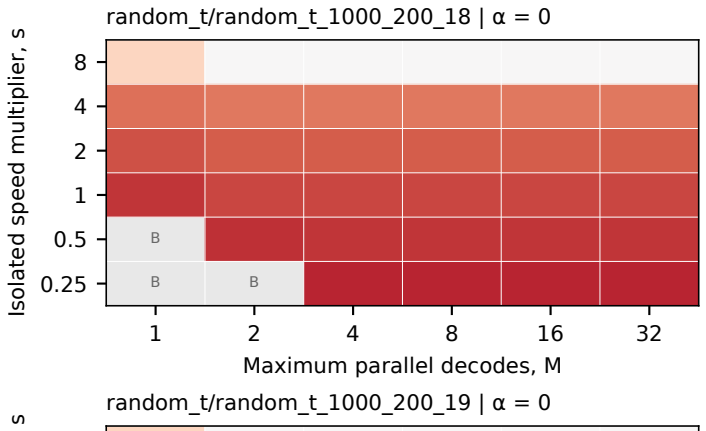
random_t/random_t_1000_200_17 | α = 0



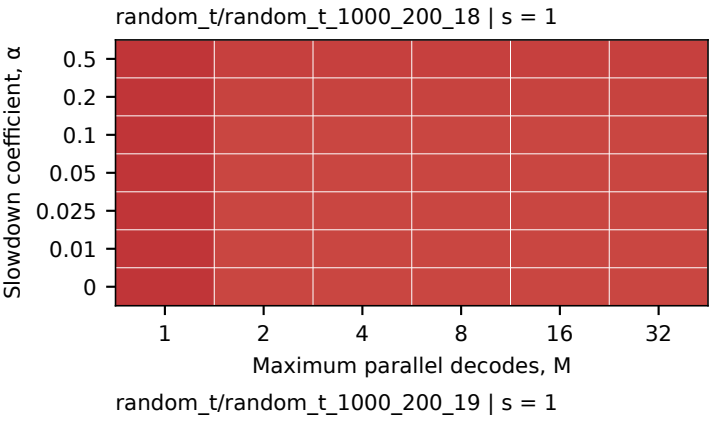
random_t/random_t_1000_200_17 | s = 1



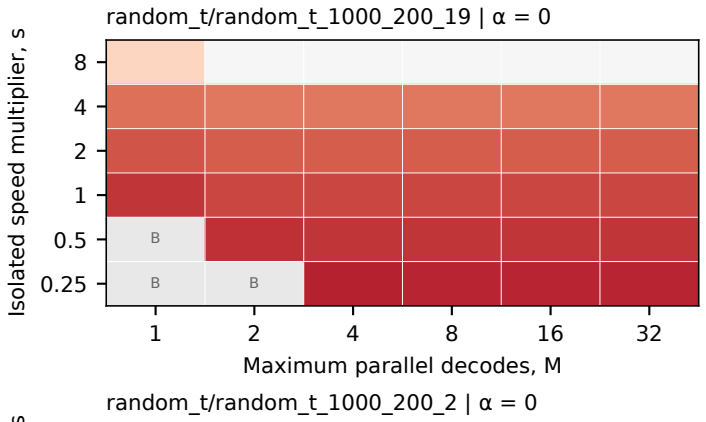
random_t/random_t_1000_200_18 | α = 0



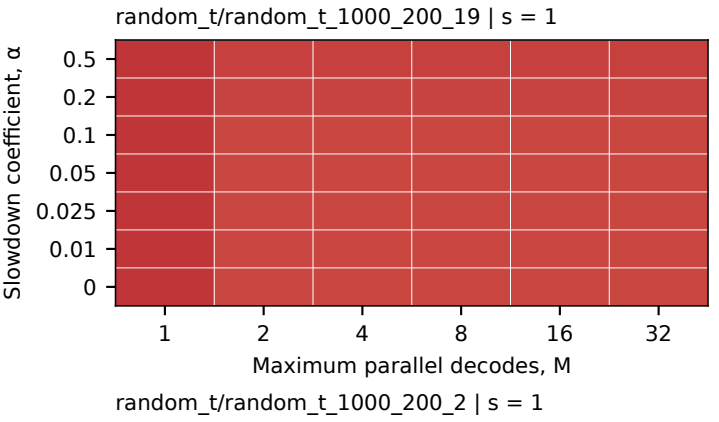
random_t/random_t_1000_200_18 | s = 1



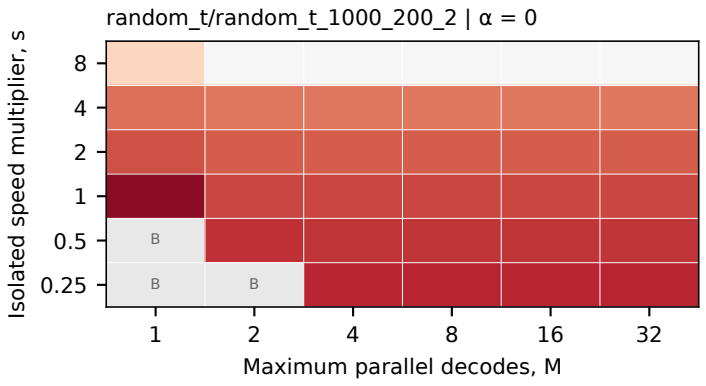
random_t/random_t_1000_200_19 | α = 0



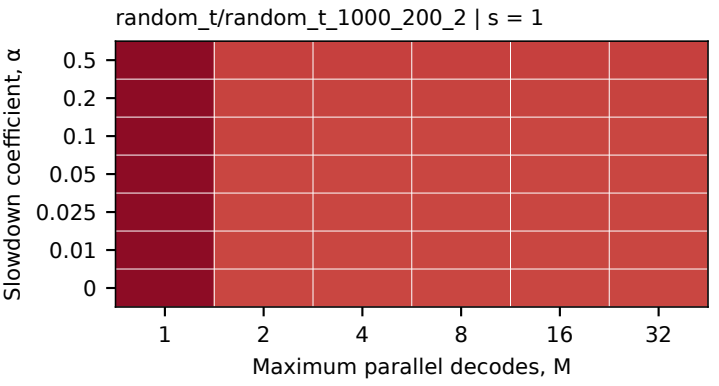
random_t/random_t_1000_200_19 | s = 1



random_t/random_t_1000_200_2 | α = 0

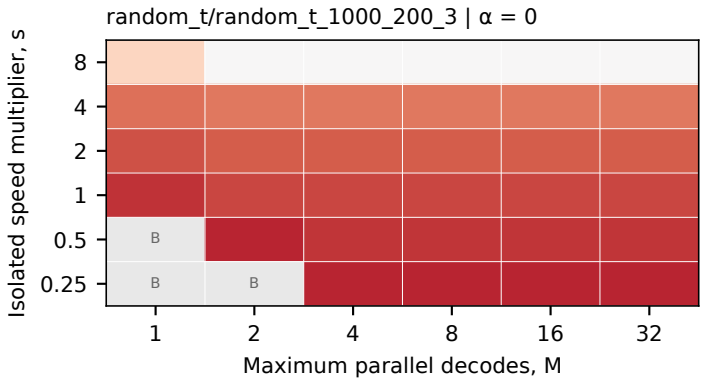


random_t/random_t_1000_200_2 | s = 1

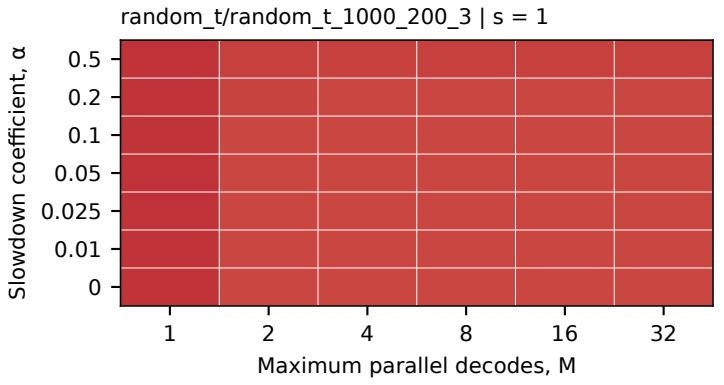


aligned · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

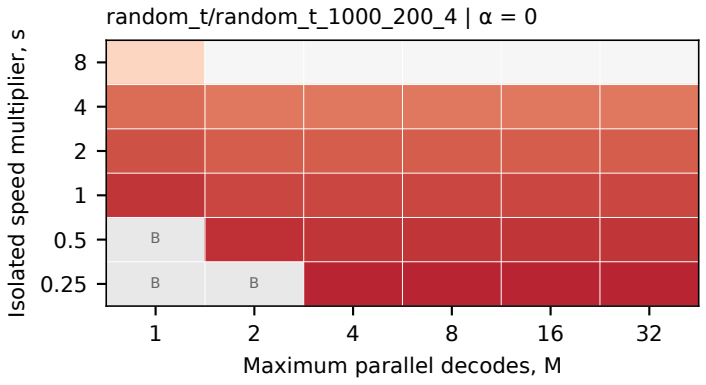
random_t/random_t_1000_200_3 | α = 0



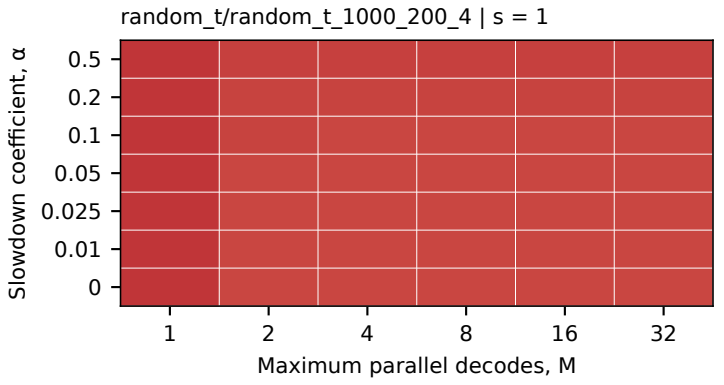
random_t/random_t_1000_200_3 | s = 1



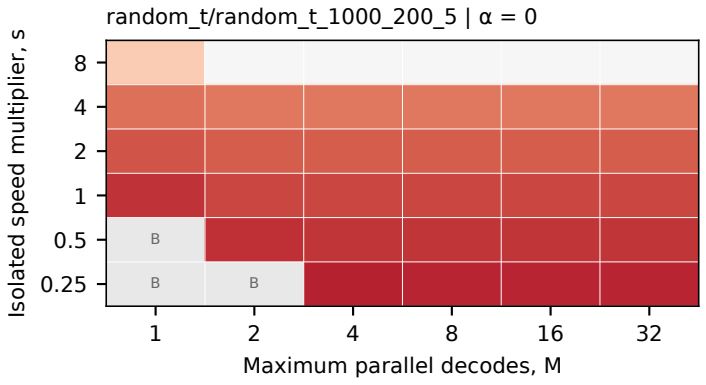
random_t/random_t_1000_200_4 | α = 0



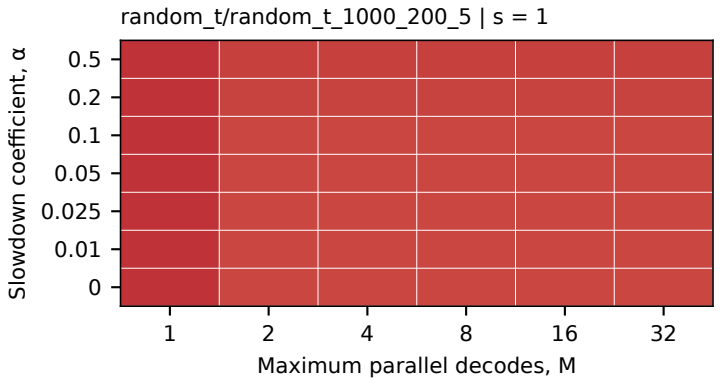
random_t/random_t_1000_200_4 | s = 1



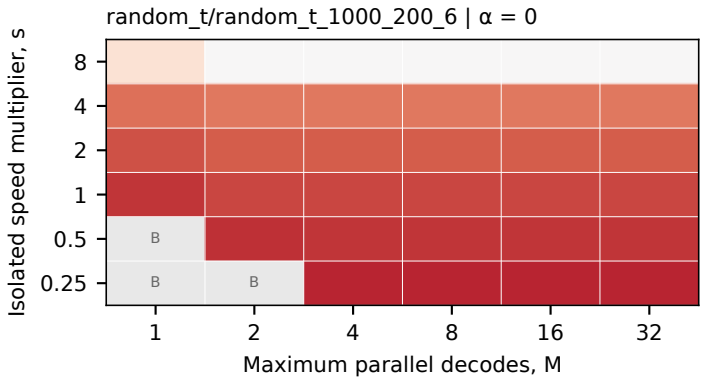
random_t/random_t_1000_200_5 | α = 0



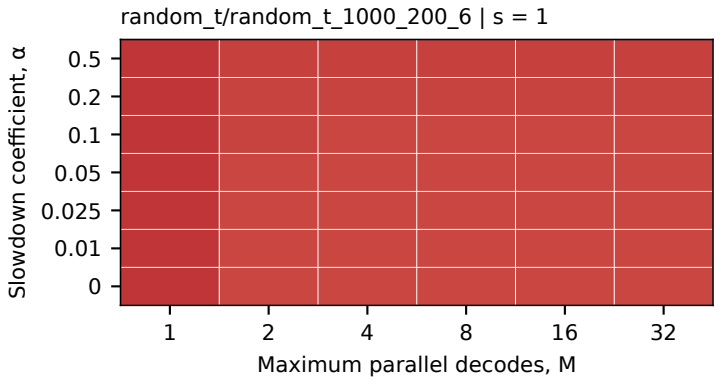
random_t/random_t_1000_200_5 | s = 1

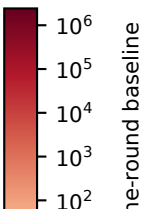


random_t/random_t_1000_200_6 | α = 0

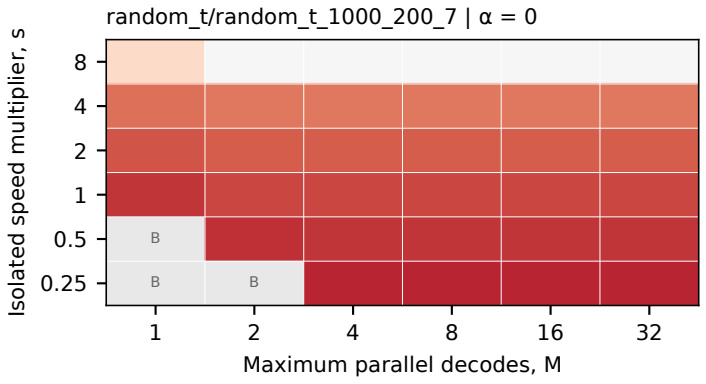


random_t/random_t_1000_200_6 | s = 1

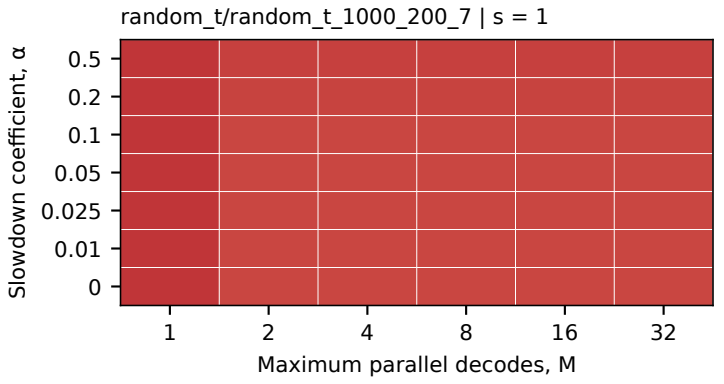




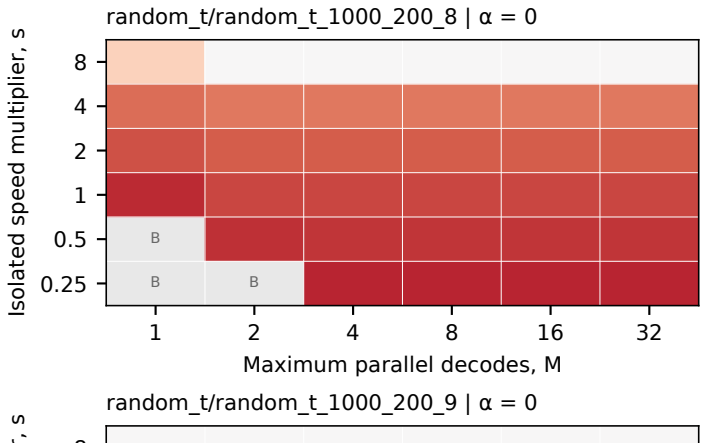
random_t/random_t_1000_200_7 | α = 0



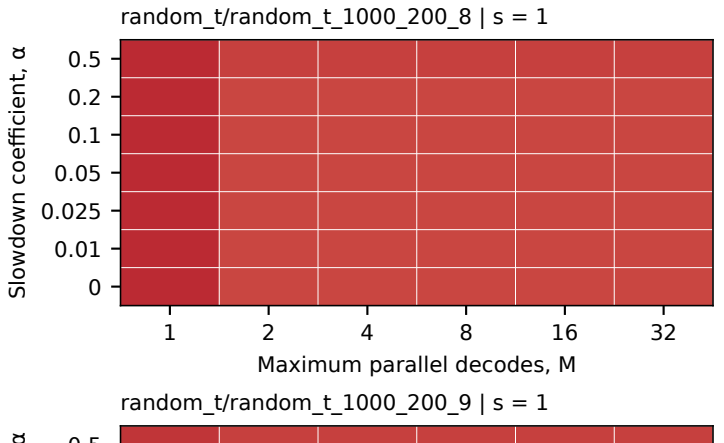
random_t/random_t_1000_200_7 | s = 1



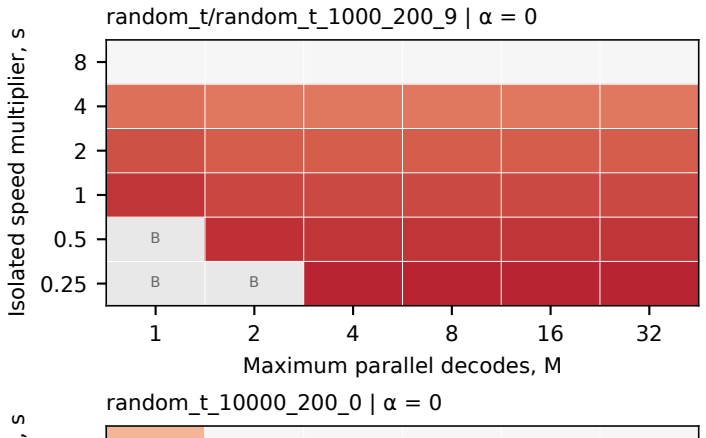
random_t/random_t_1000_200_8 | α = 0



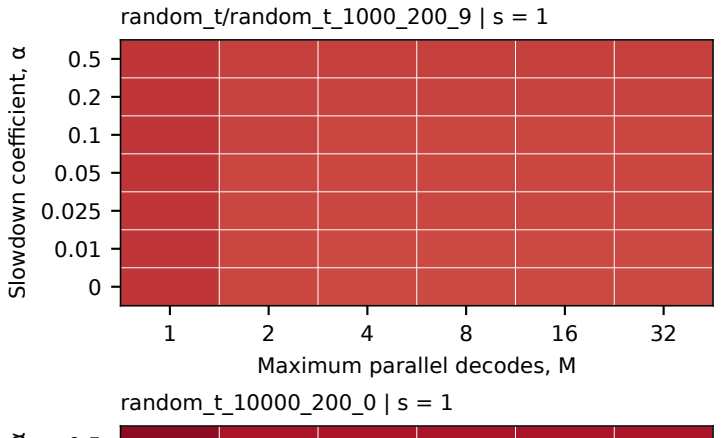
random_t/random_t_1000_200_8 | s = 1



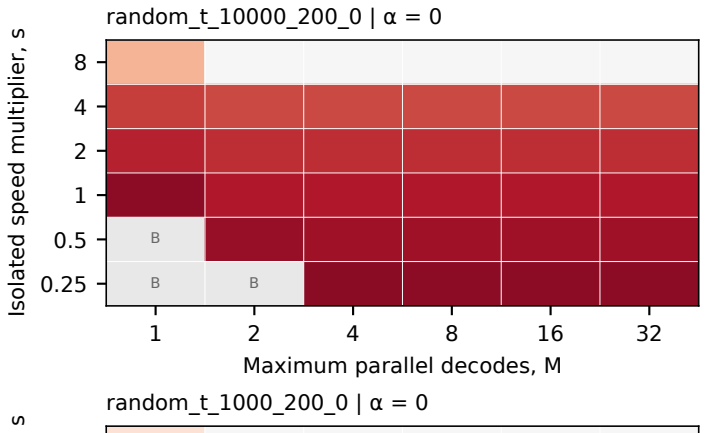
random_t/random_t_1000_200_9 | α = 0



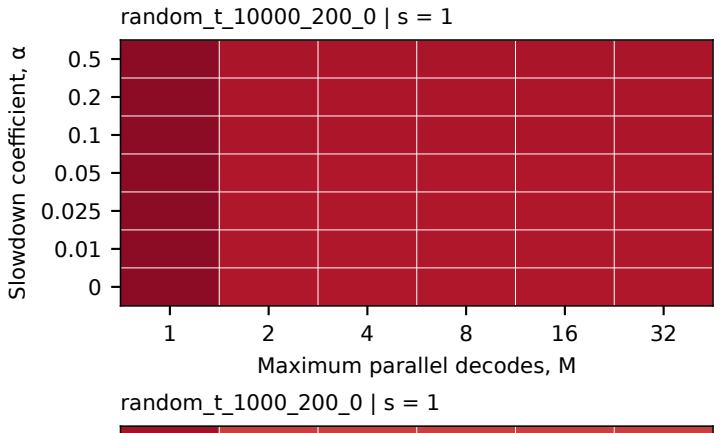
random_t/random_t_1000_200_9 | s = 1



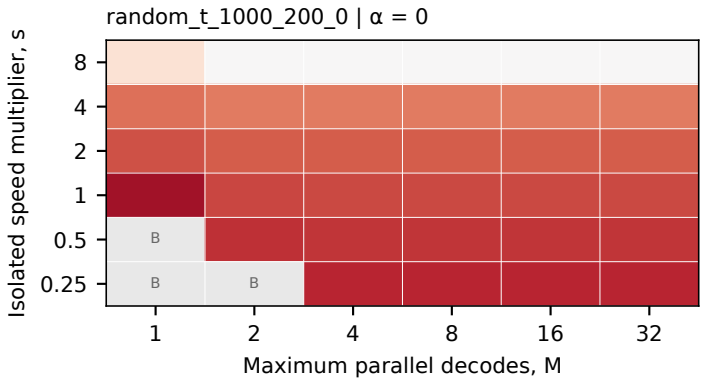
random_t_10000_200_0 | α = 0



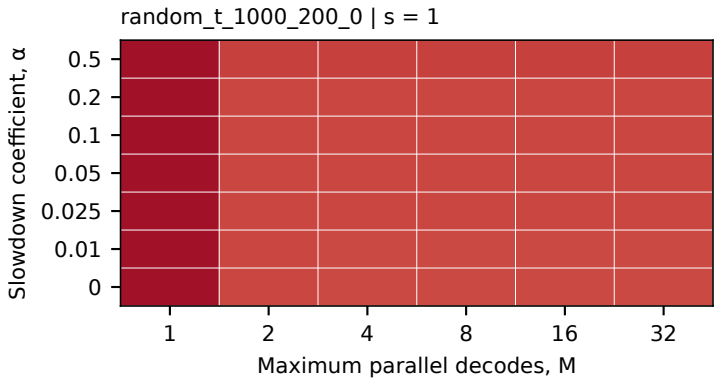
random_t_10000_200_0 | s = 1



random_t_1000_200_0 | α = 0

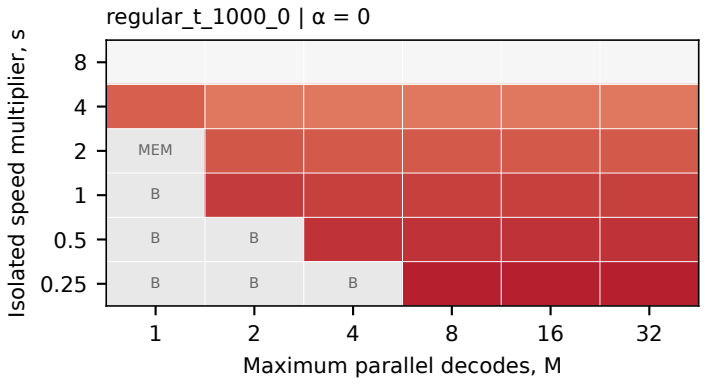


random_t_1000_200_0 | s = 1

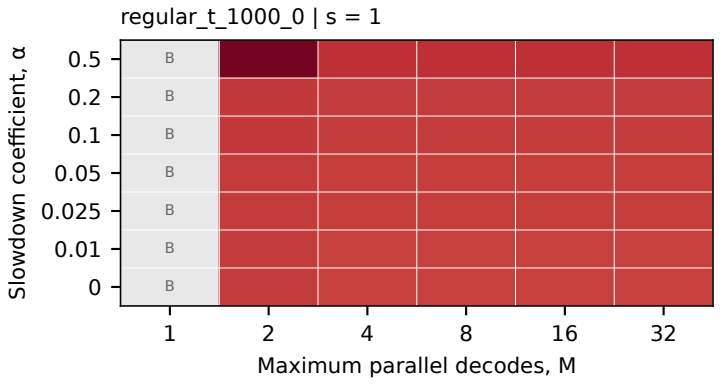


aligned · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

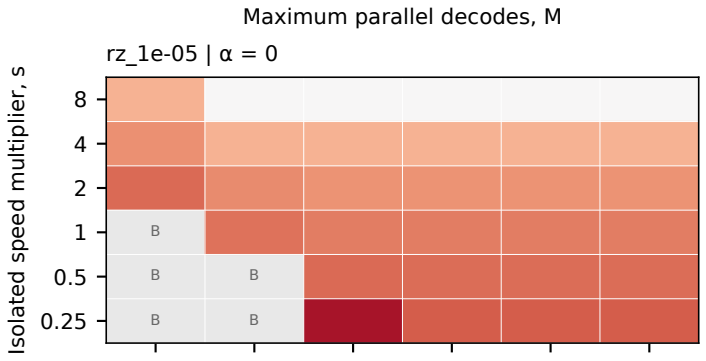
regular_t_1000_0 | α = 0



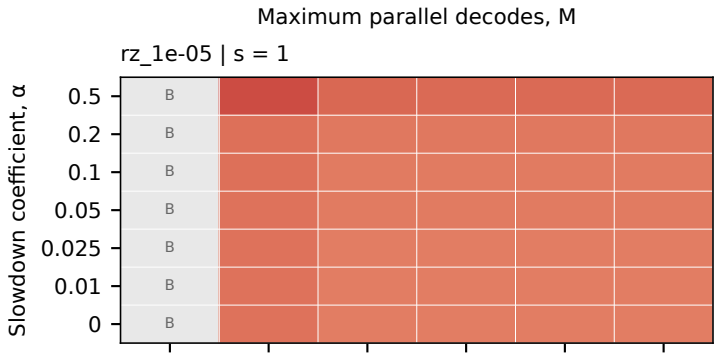
regular_t_1000_0 | s = 1



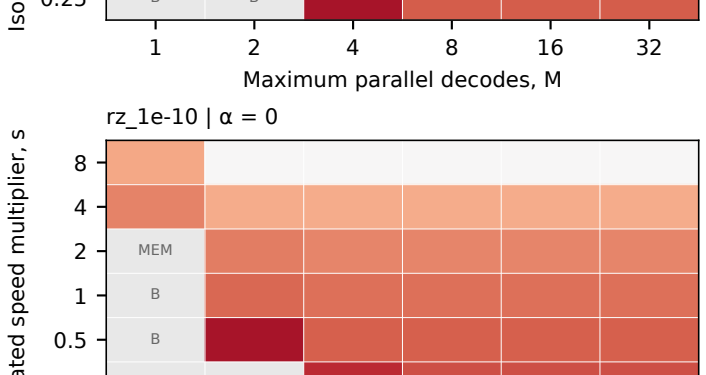
rz_1e-05 | α = 0



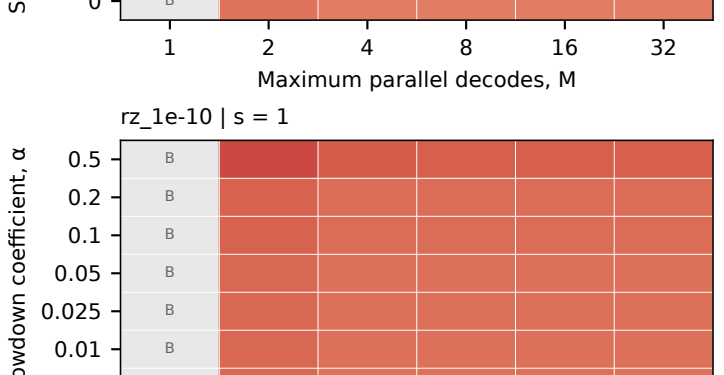
rz_1e-05 | s = 1



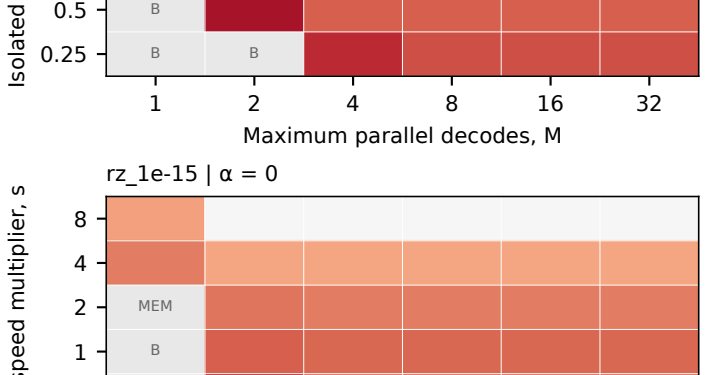
rz_1e-10 | α = 0



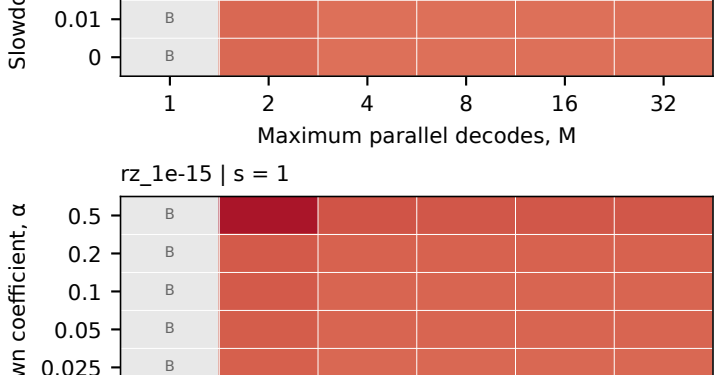
rz_1e-10 | s = 1



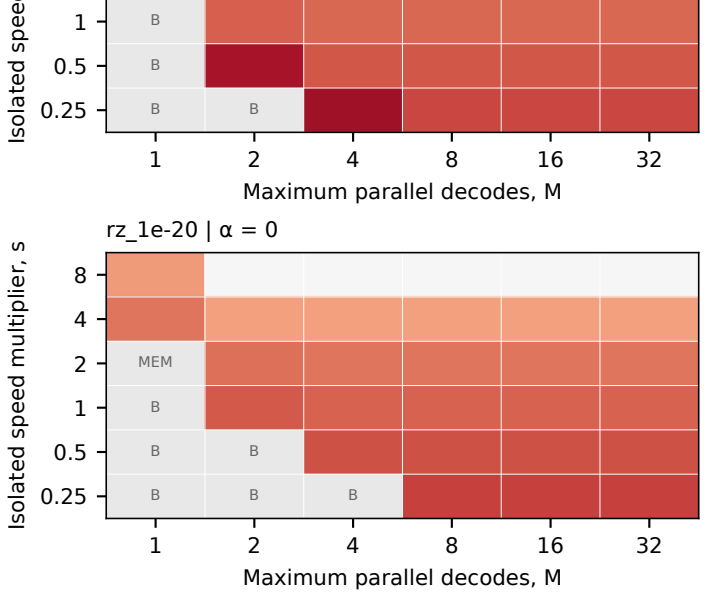
rz_1e-15 | α = 0



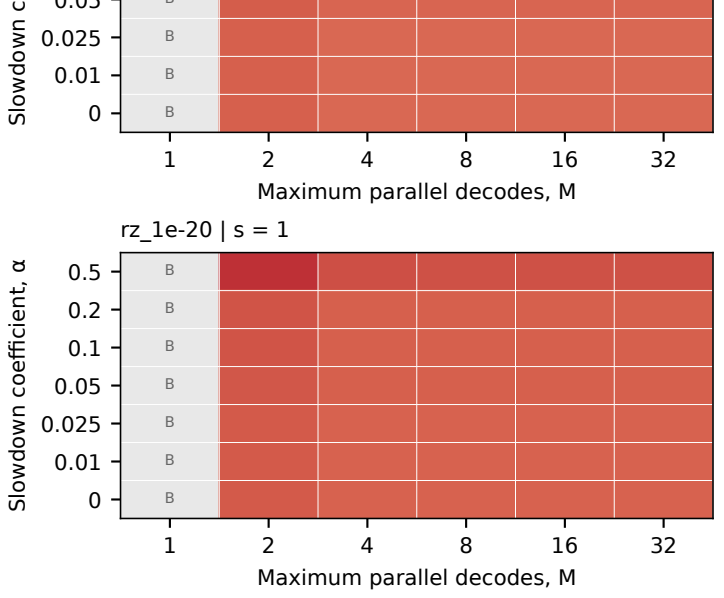
rz_1e-15 | s = 1

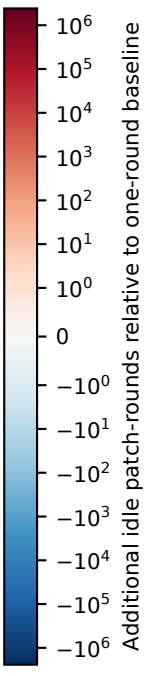


rz_1e-20 | α = 0

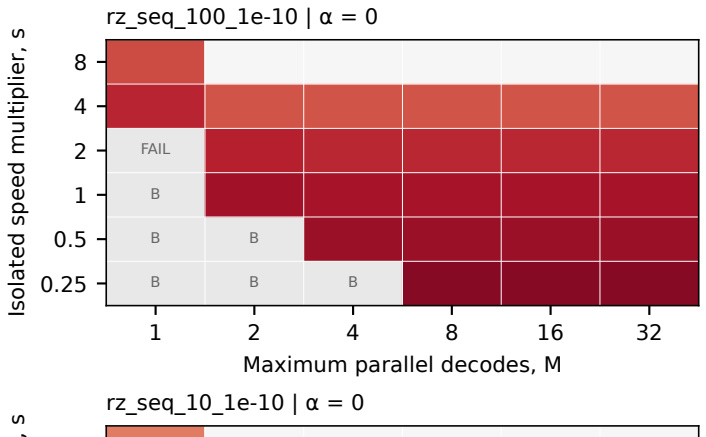


rz_1e-20 | s = 1

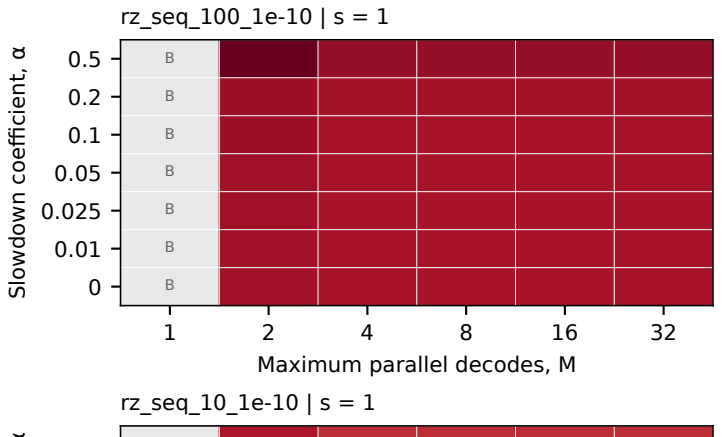




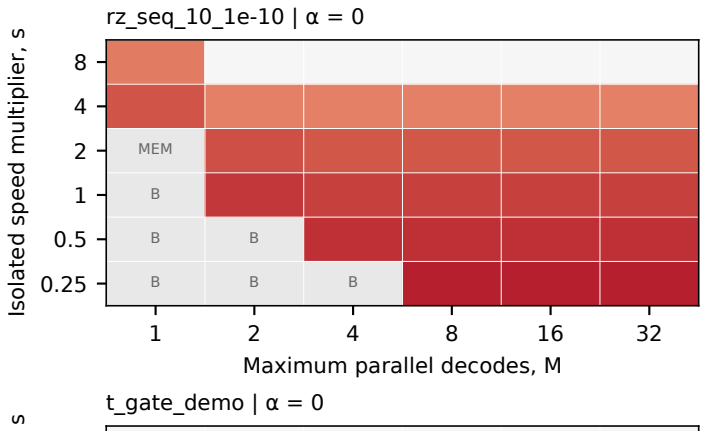
rz_seq_100_1e-10 | α = 0



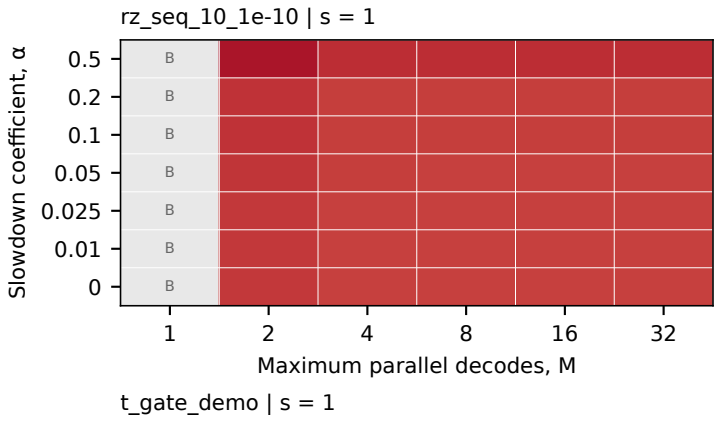
rz_seq_100_1e-10 | s = 1



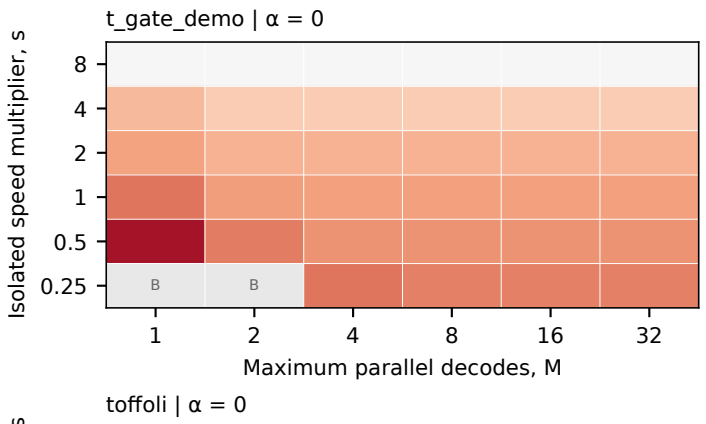
rz_seq_10_1e-10 | α = 0



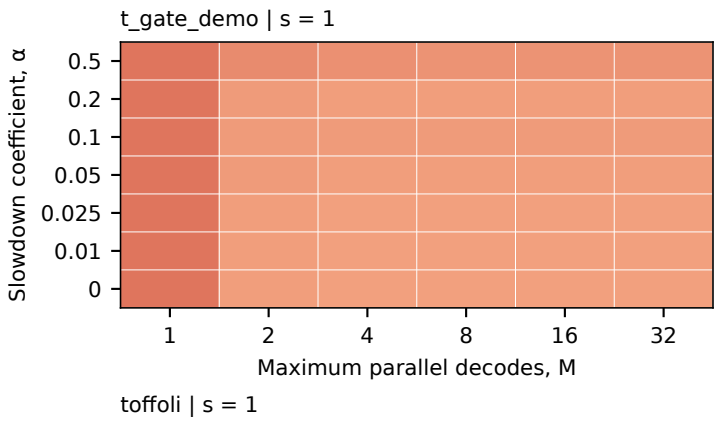
rz_seq_10_1e-10 | s = 1



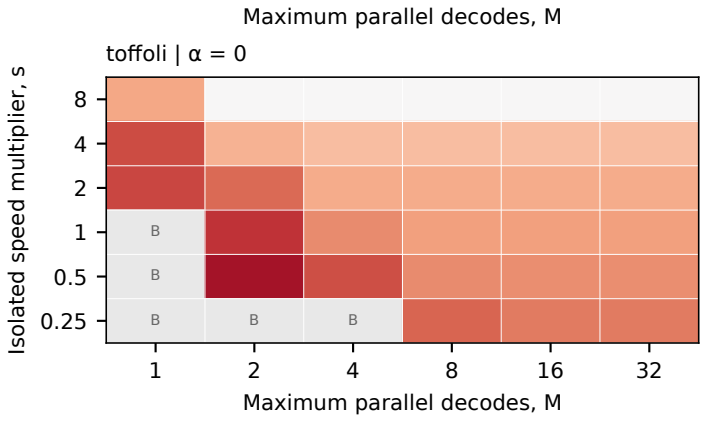
t_gate_demo | α = 0



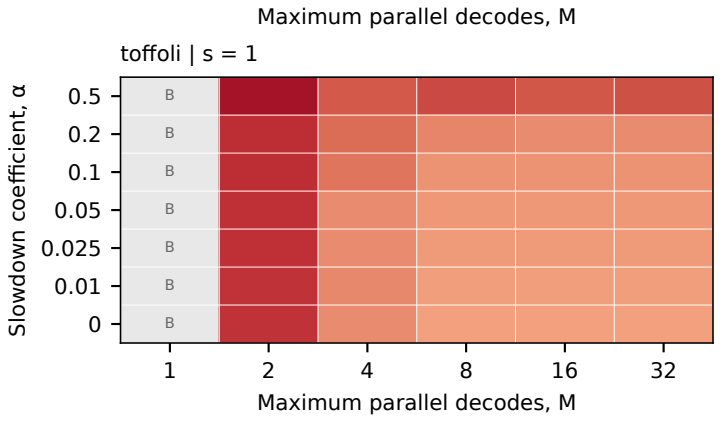
t_gate_demo | s = 1



toffoli | α = 0

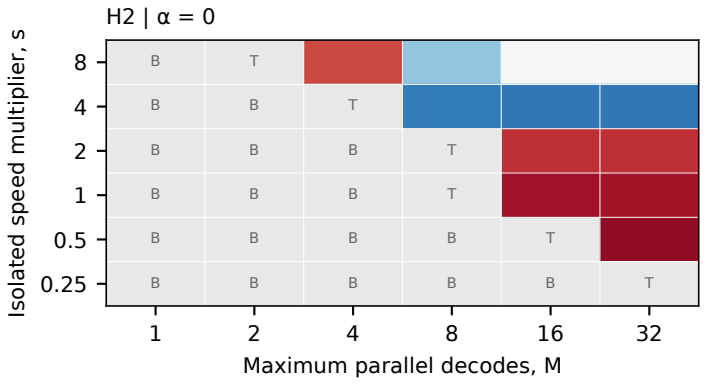


toffoli | s = 1

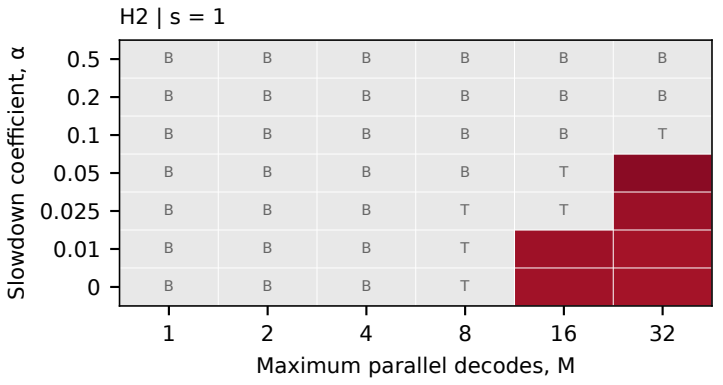


parallel · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

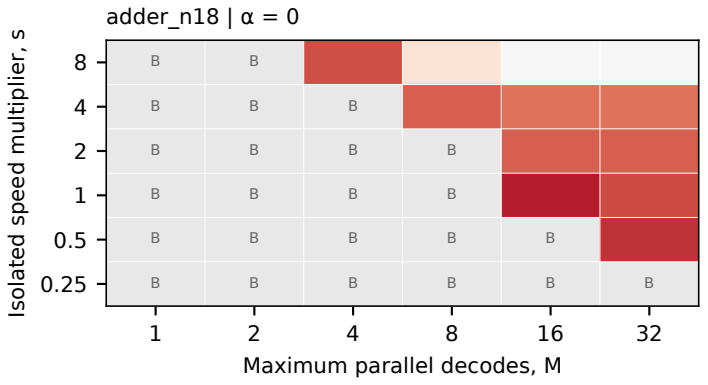
H2 | α = 0



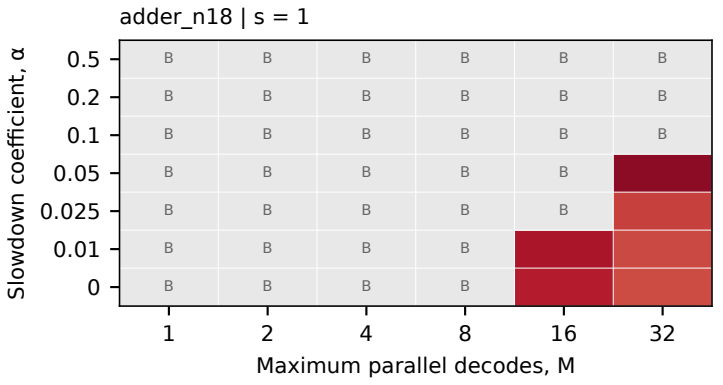
H2 | s = 1



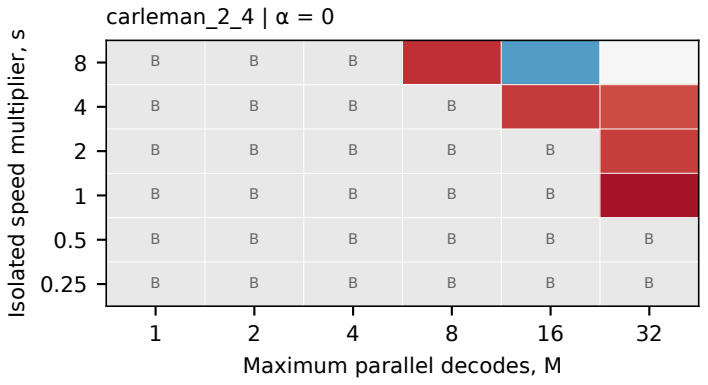
adder_n18 | α = 0



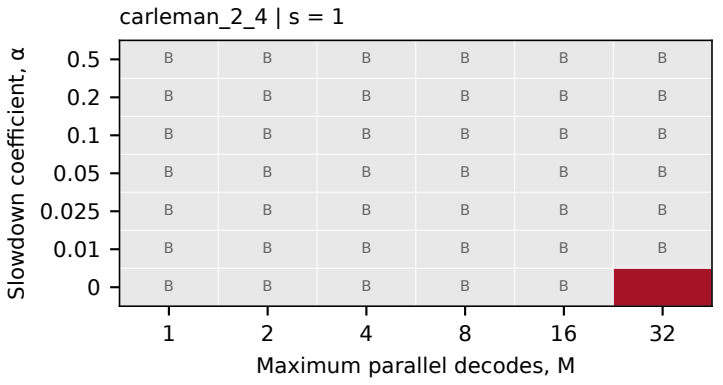
adder_n18 | s = 1



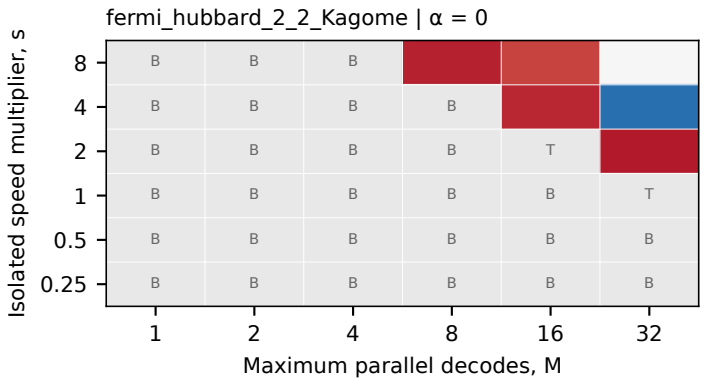
carleman_2_4 | α = 0



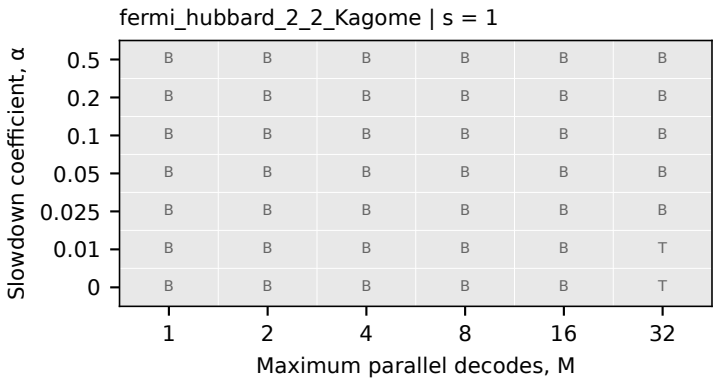
carleman_2_4 | s = 1



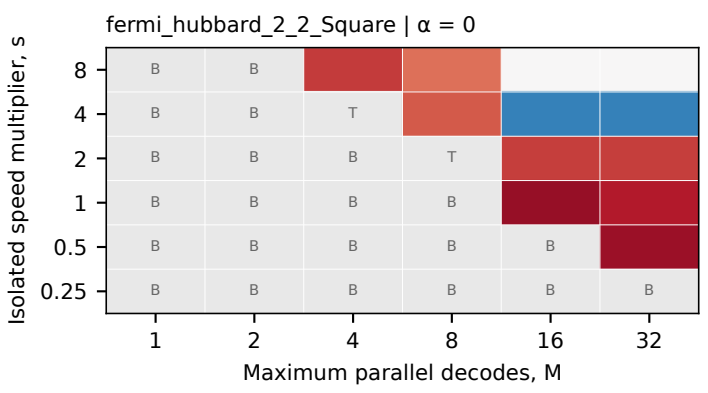
fermi_hubbard_2_2_Kagome | α = 0



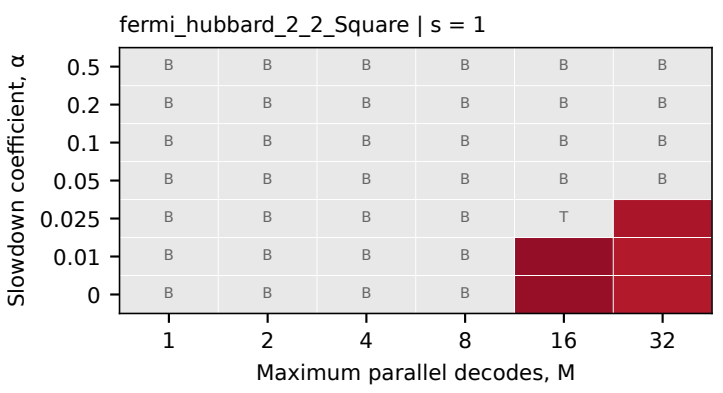
fermi_hubbard_2_2_Kagome | s = 1



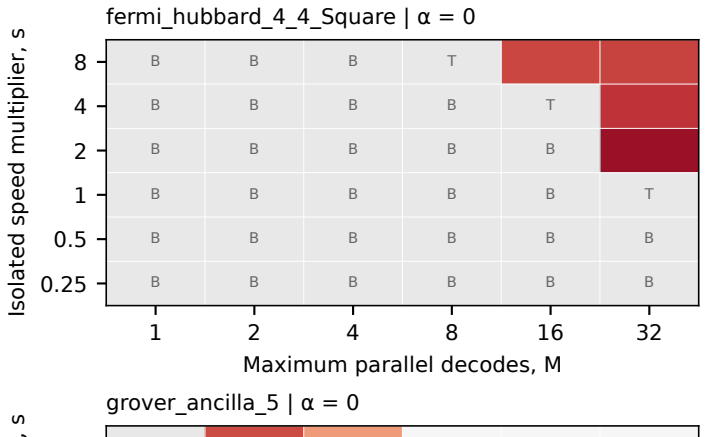
fermi_hubbard_2_2_Square | α = 0



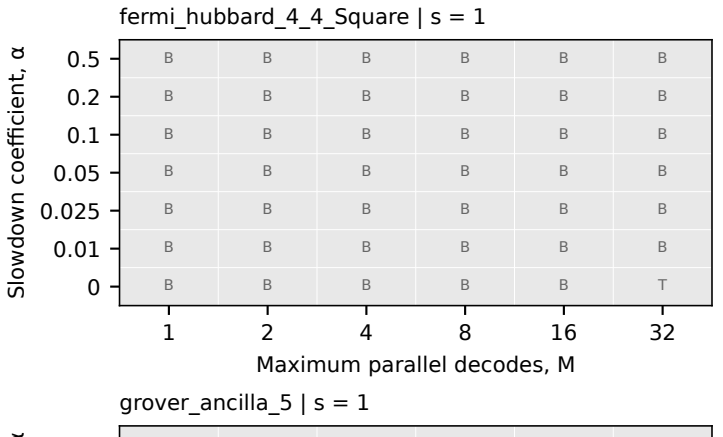
fermi_hubbard_2_2_Square | s = 1



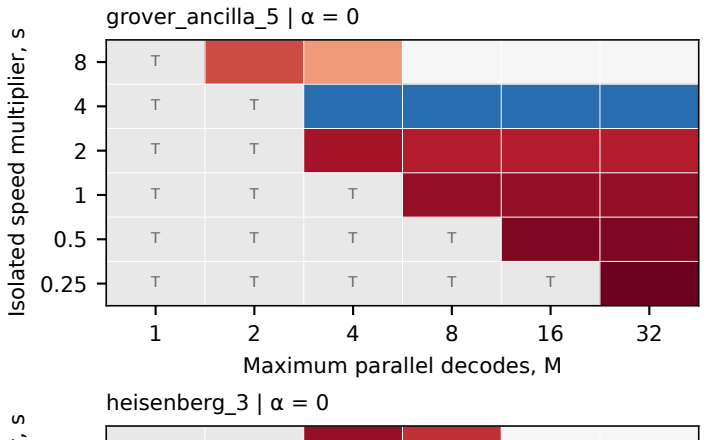
fermi_hubbard_4_4_Square | α = 0



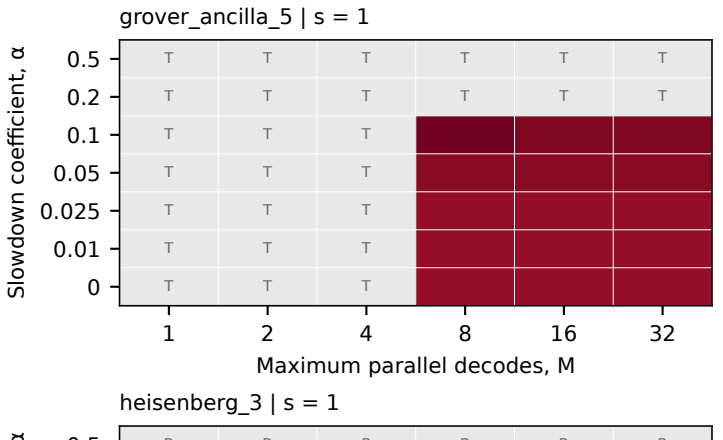
fermi_hubbard_4_4_Square | s = 1



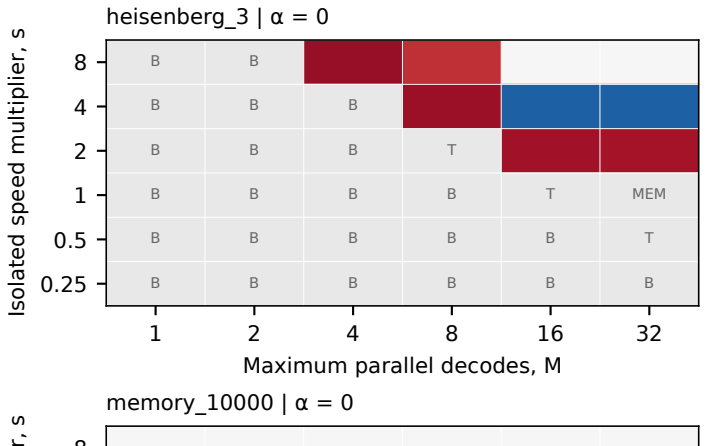
grover_ancilla_5 | α = 0



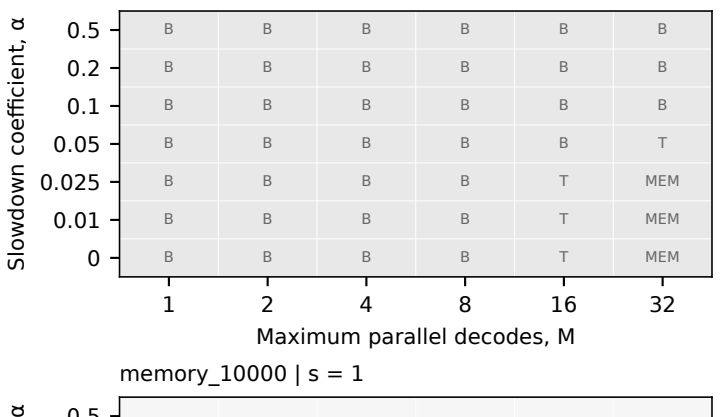
grover_ancilla_5 | s = 1



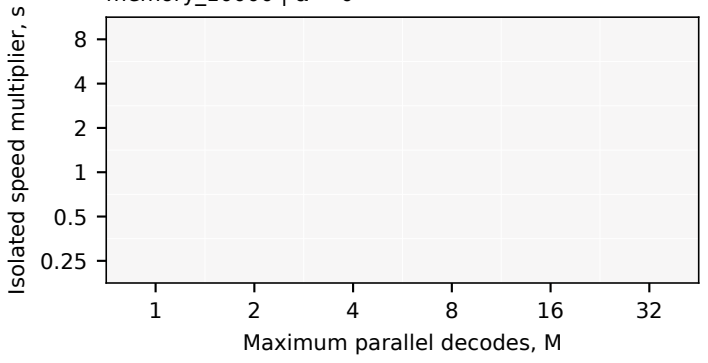
heisenberg_3 | α = 0



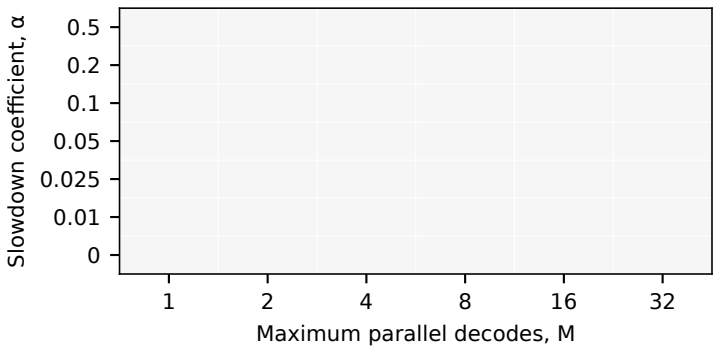
heisenberg_3 | s = 1

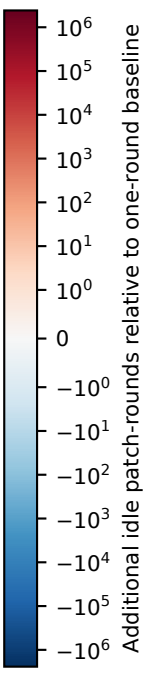


memory_10000 | α = 0



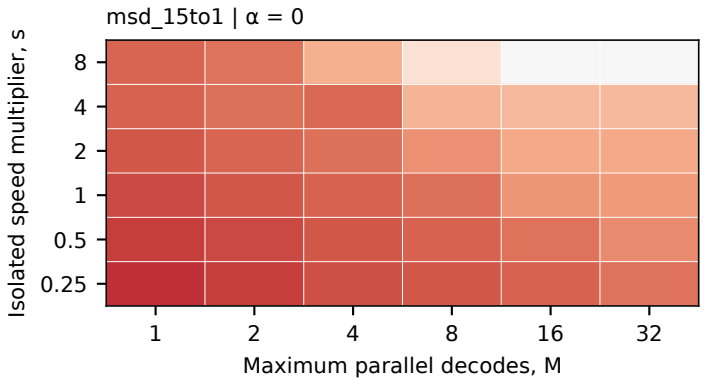
memory_10000 | s = 1



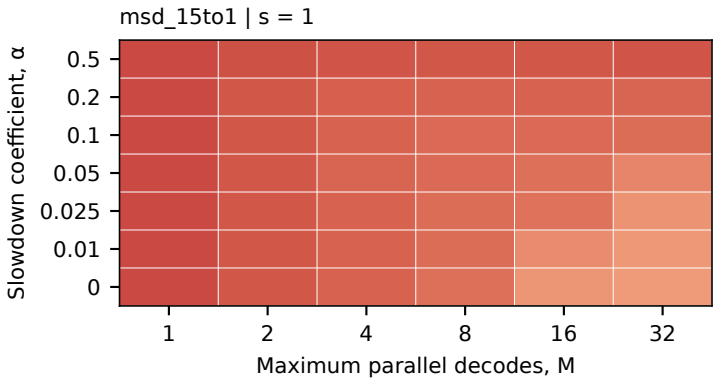


parallel · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

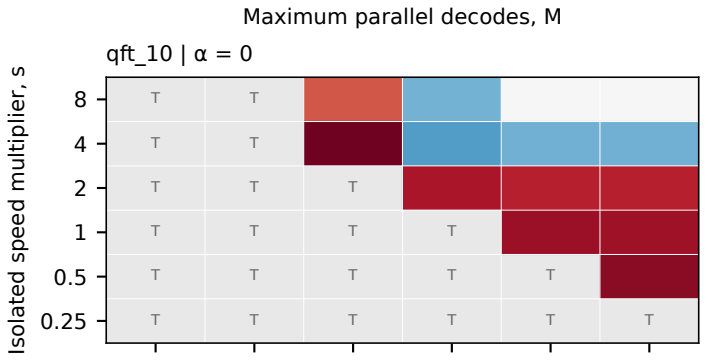
msd_15to1 | α = 0



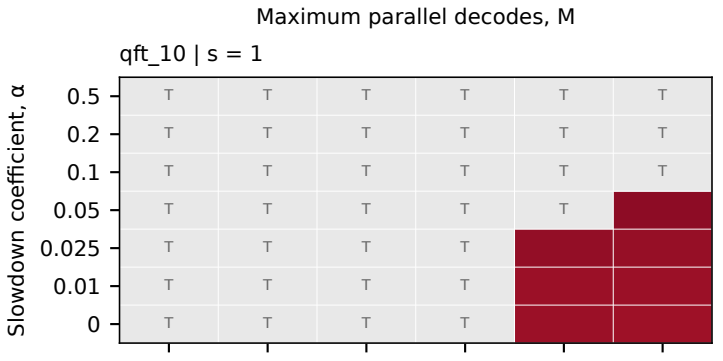
msd_15to1 | s = 1



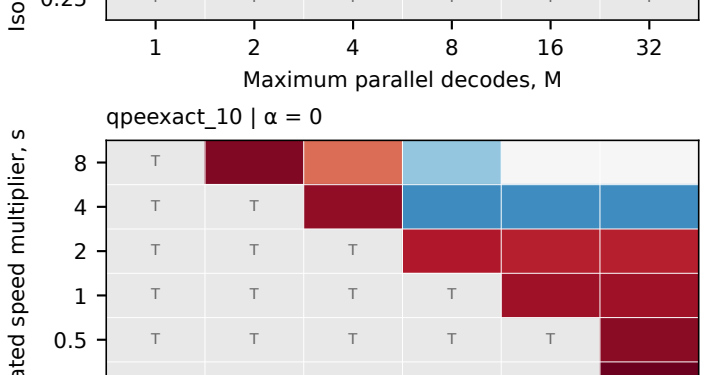
qft_10 | α = 0



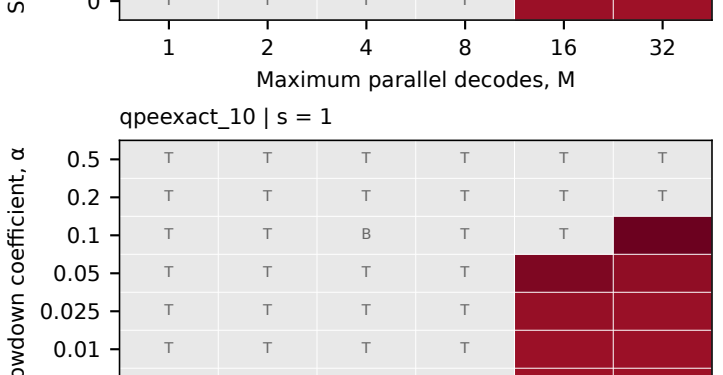
qft_10 | s = 1



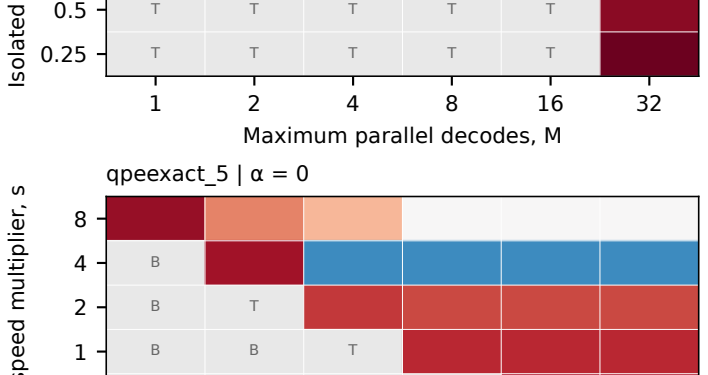
qpeexact_10 | α = 0



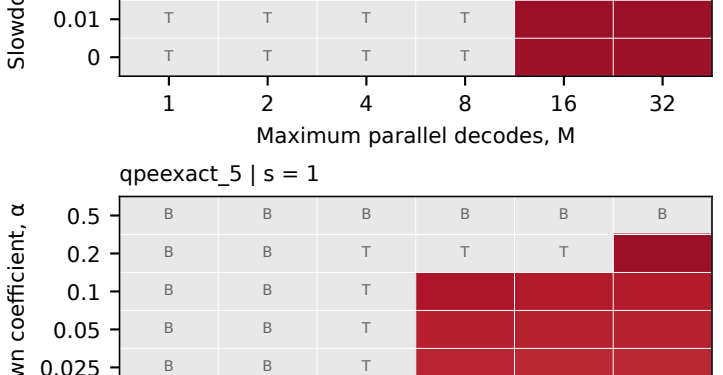
qpeexact_10 | s = 1



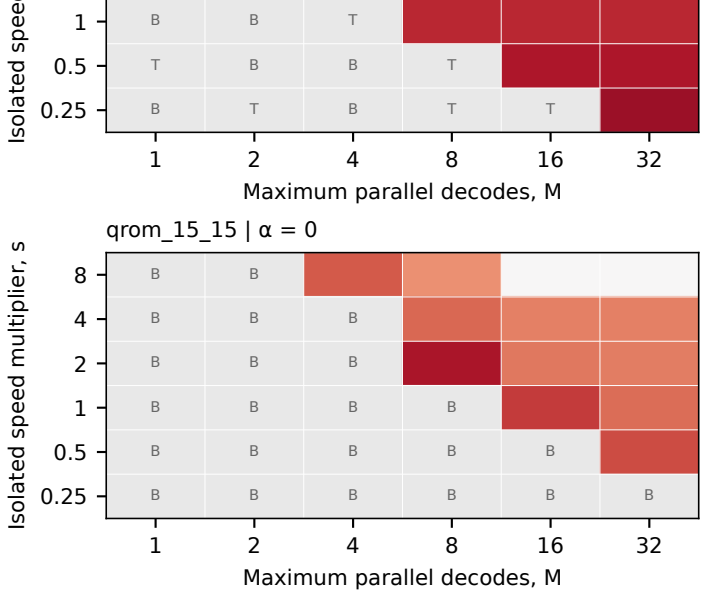
qpeexact_5 | α = 0



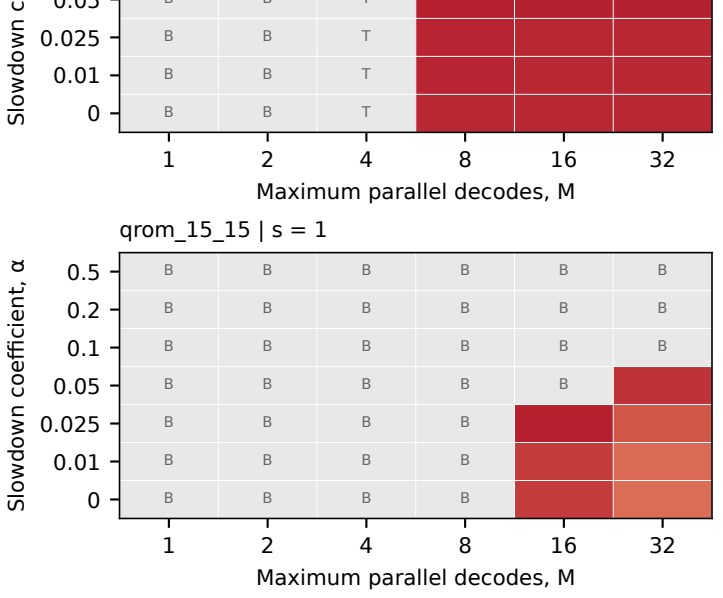
qpeexact_5 | s = 1

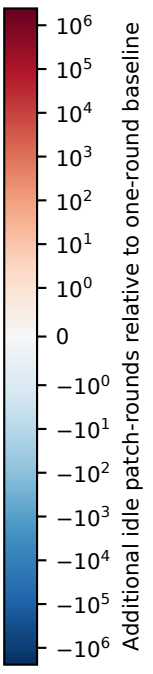


qrom_15_15 | α = 0

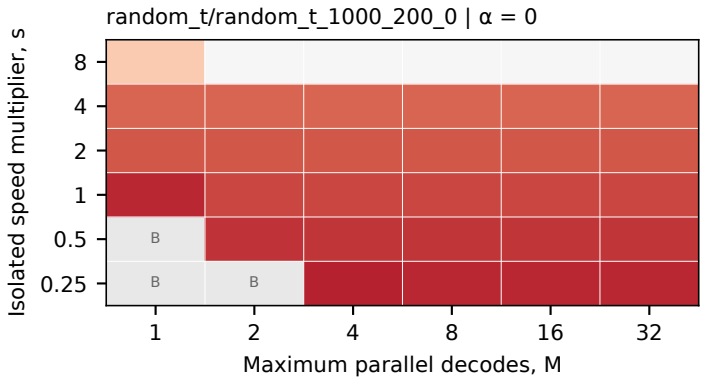


qrom_15_15 | s = 1

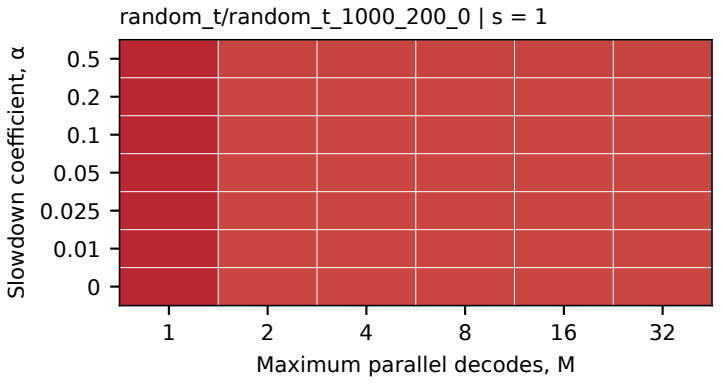




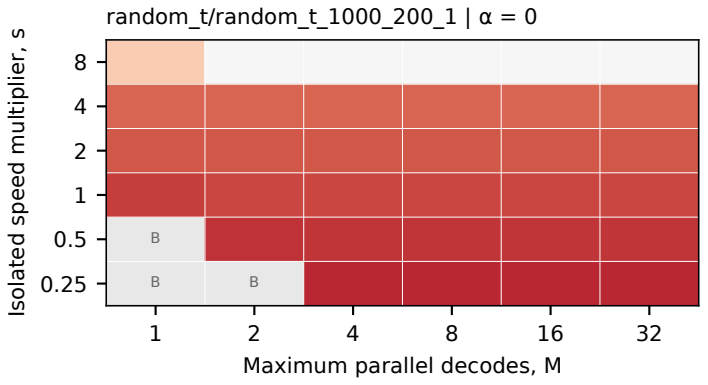
random_t/random_t_1000_200_0 | α = 0



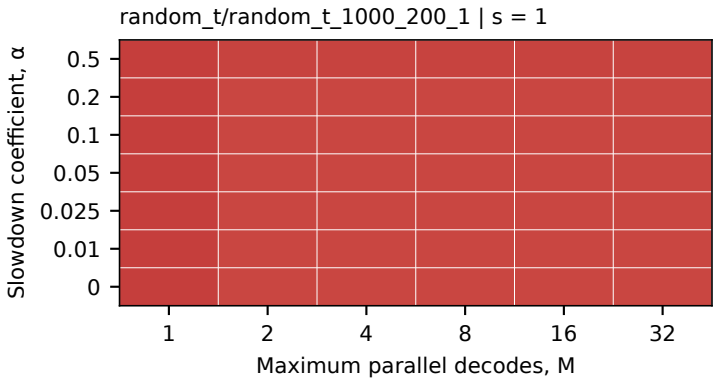
random_t/random_t_1000_200_0 | s = 1



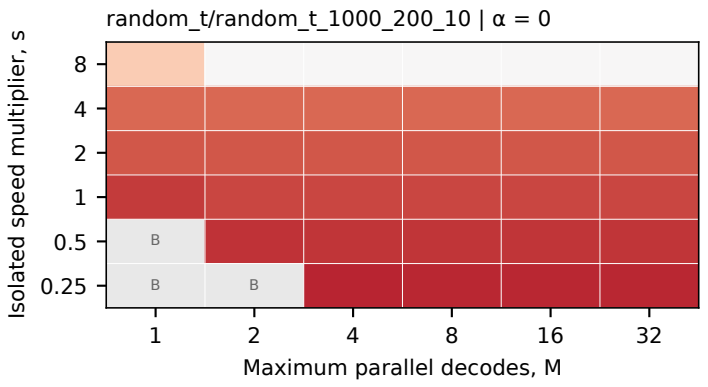
random_t/random_t_1000_200_1 | α = 0



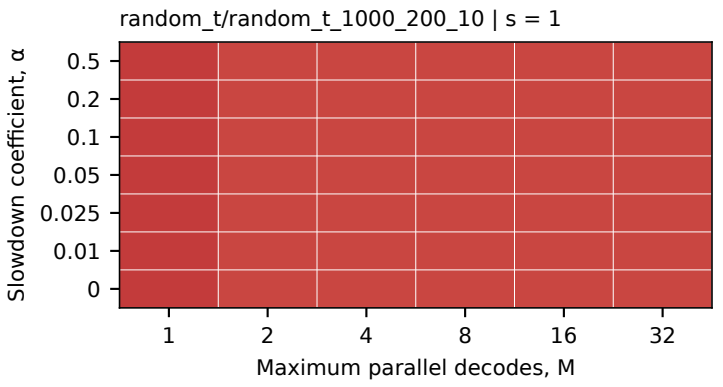
random_t/random_t_1000_200_1 | s = 1



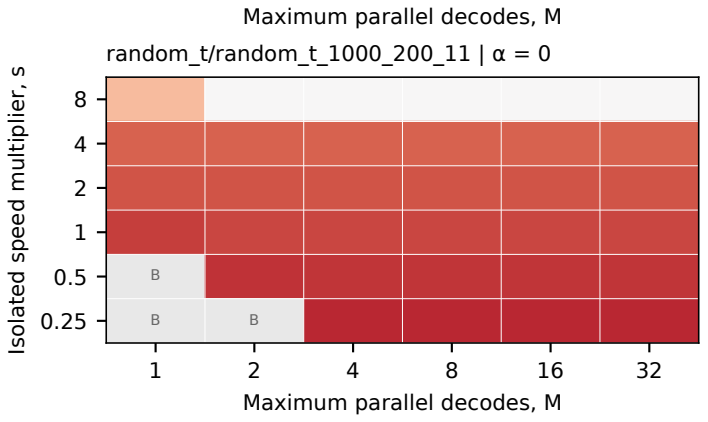
random_t/random_t_1000_200_10 | α = 0



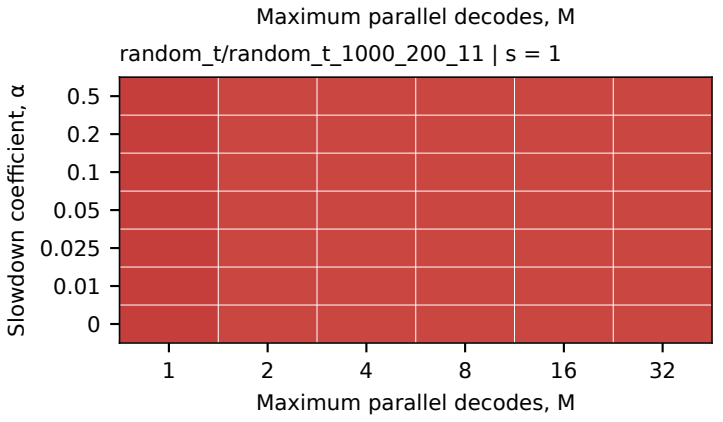
random_t/random_t_1000_200_10 | s = 1



random_t/random_t_1000_200_11 | α = 0

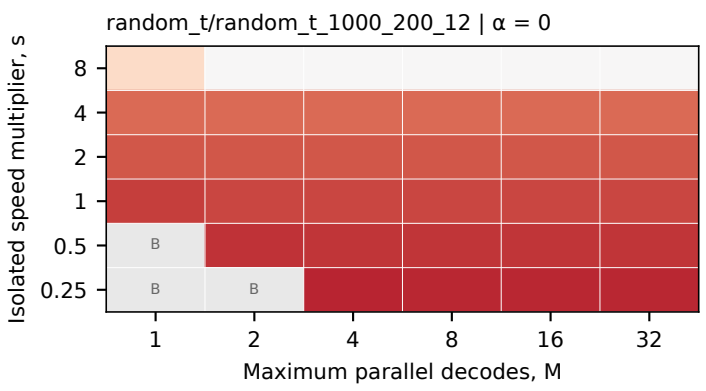


random_t/random_t_1000_200_11 | s = 1

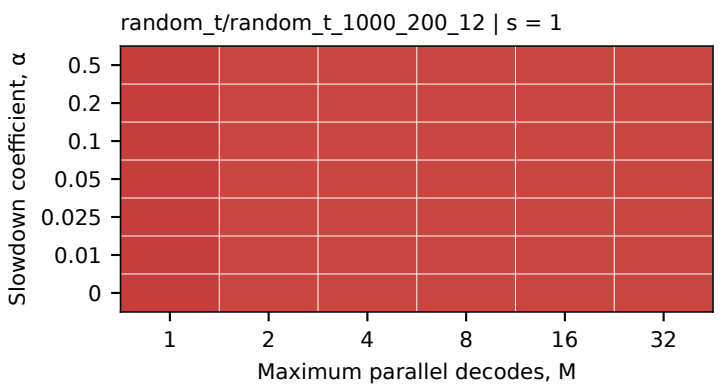


parallel · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

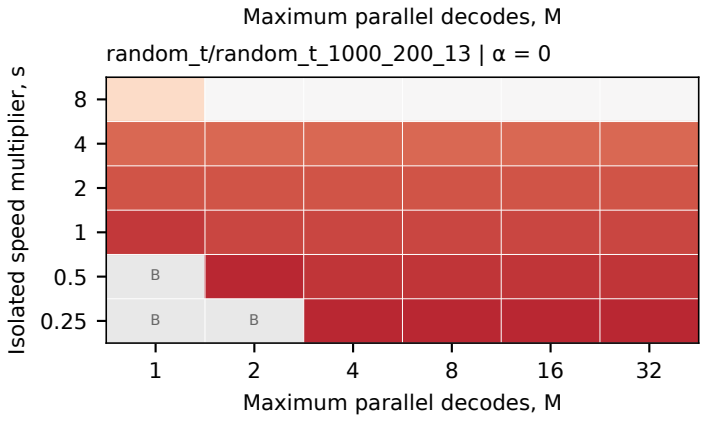
random_t/random_t_1000_200_12 | α = 0



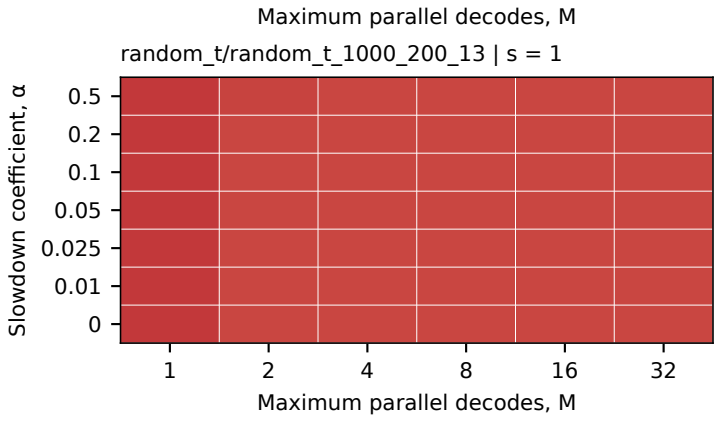
random_t/random_t_1000_200_12 | s = 1



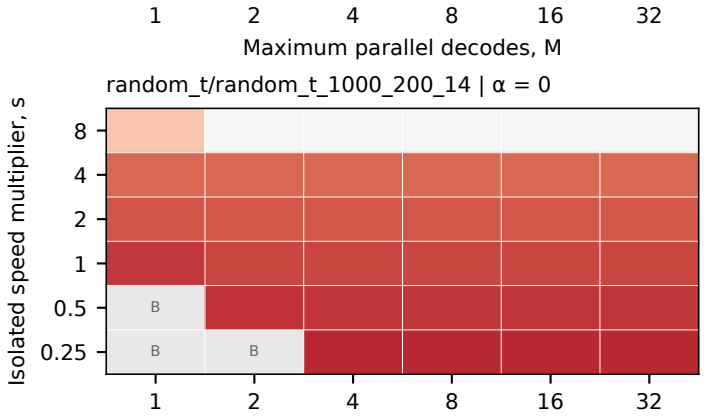
random_t/random_t_1000_200_13 | α = 0



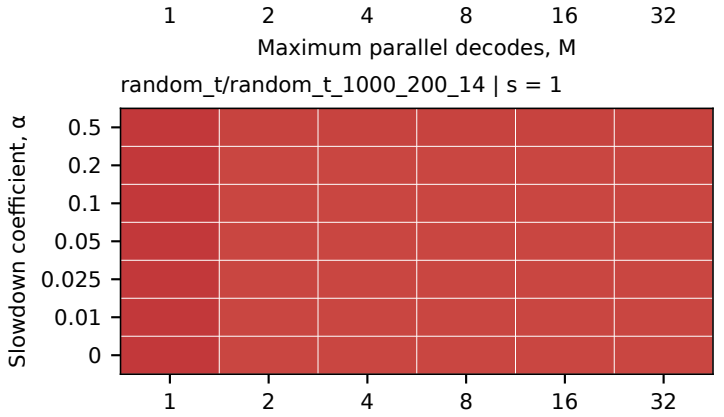
random_t/random_t_1000_200_13 | s = 1



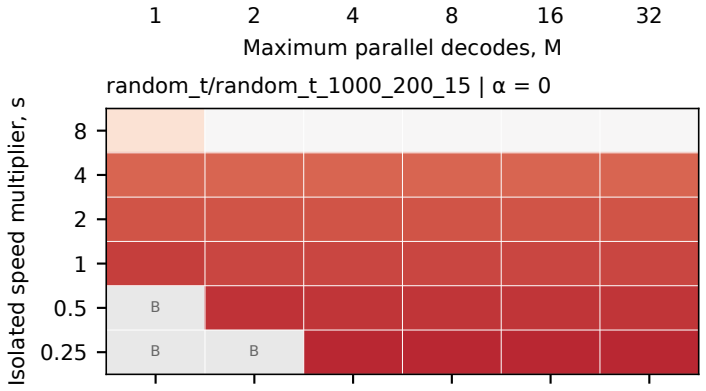
random_t/random_t_1000_200_14 | α = 0



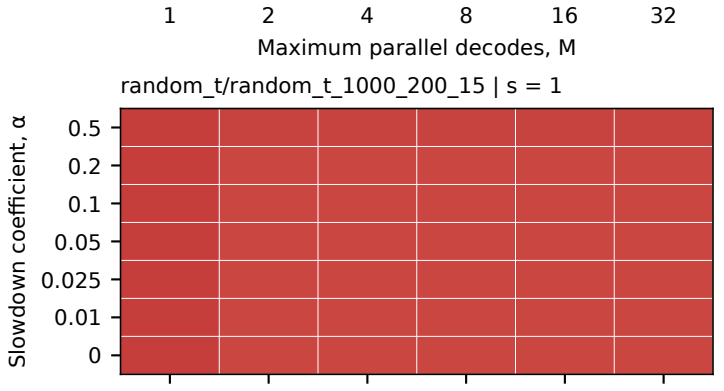
random_t/random_t_1000_200_14 | s = 1

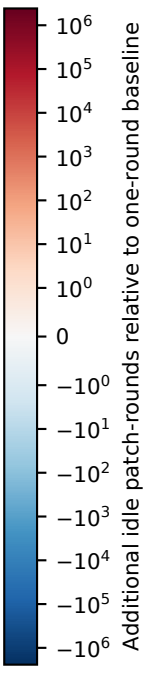


random_t/random_t_1000_200_15 | α = 0

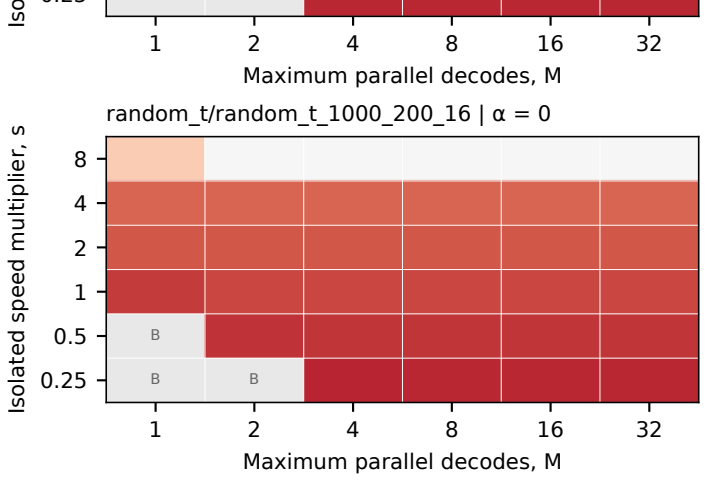


random_t/random_t_1000_200_15 | s = 1

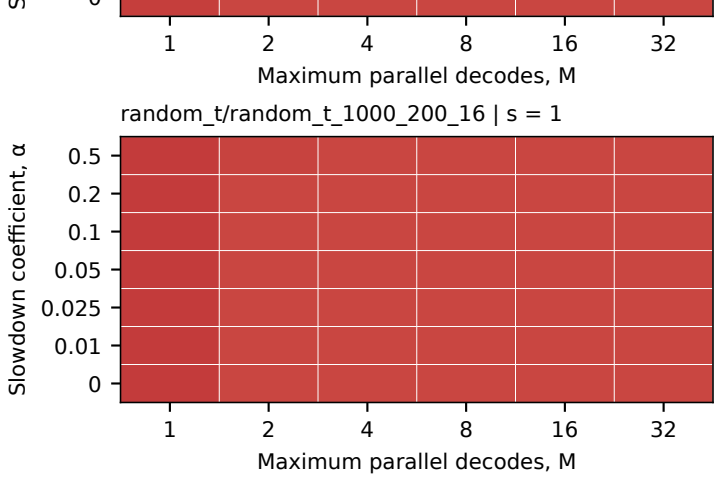




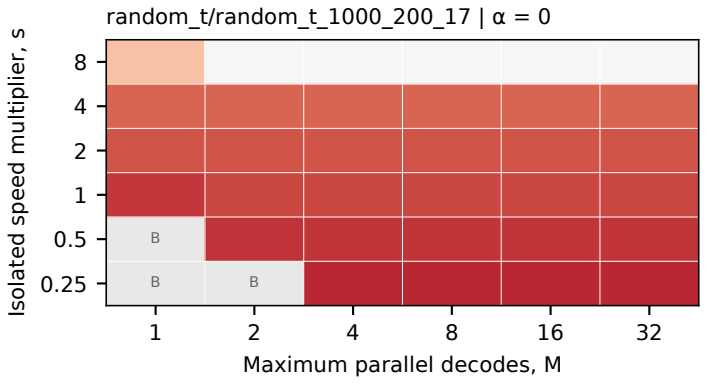
random_t/random_t_1000_200_16 | α = 0



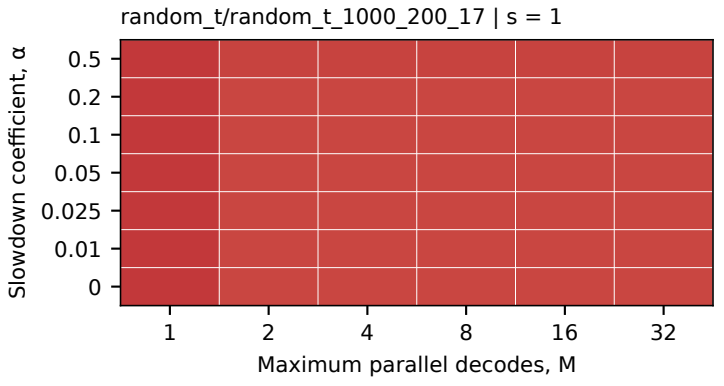
random_t/random_t_1000_200_16 | s = 1



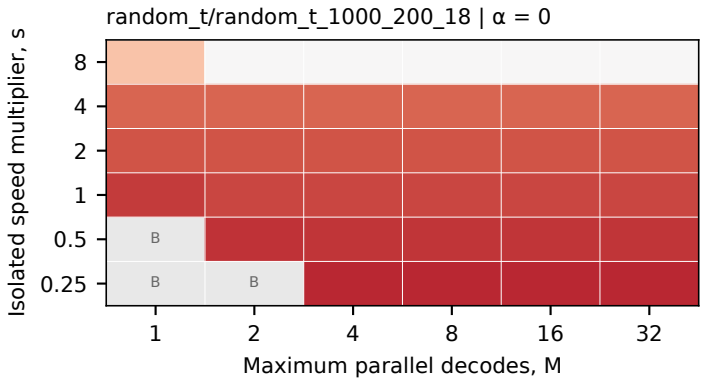
random_t/random_t_1000_200_17 | α = 0



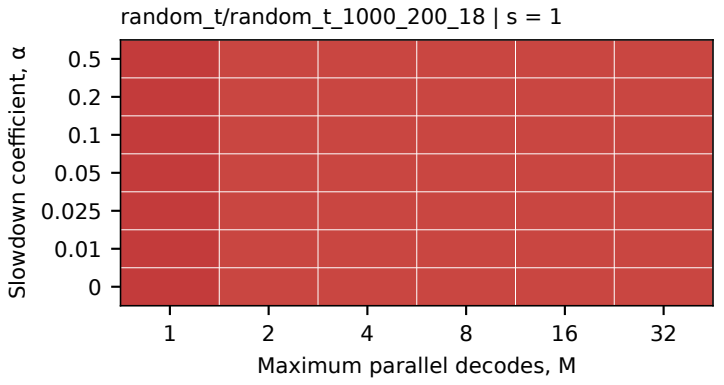
random_t/random_t_1000_200_17 | s = 1



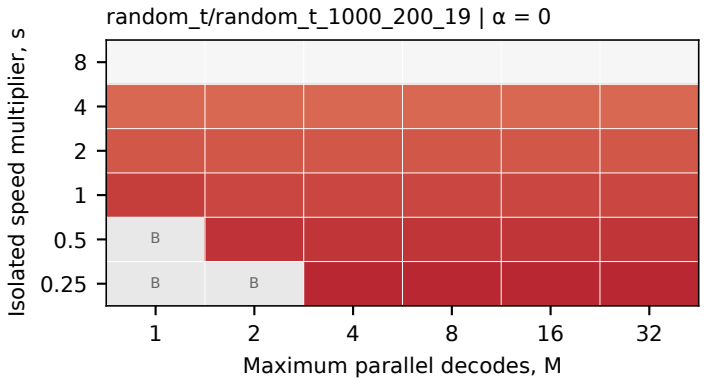
random_t/random_t_1000_200_18 | α = 0



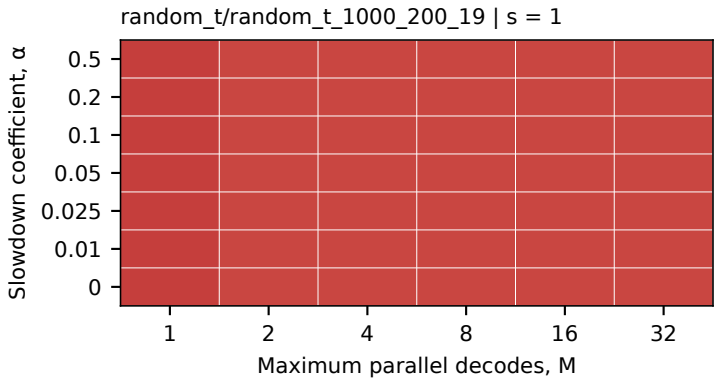
random_t/random_t_1000_200_18 | s = 1



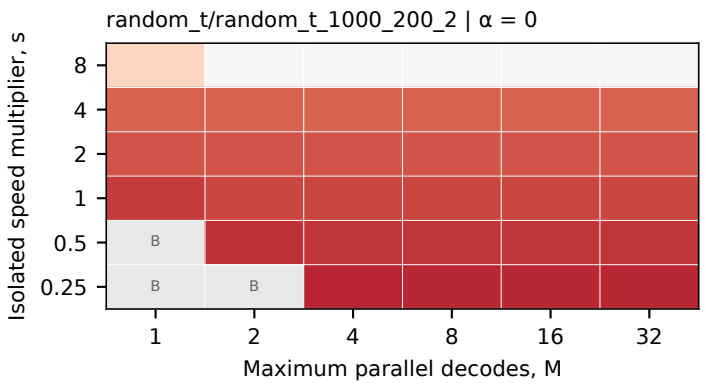
random_t/random_t_1000_200_19 | α = 0



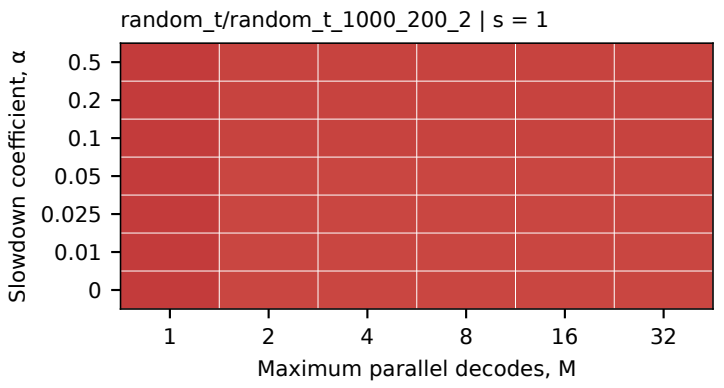
random_t/random_t_1000_200_19 | s = 1



random_t/random_t_1000_200_2 | α = 0

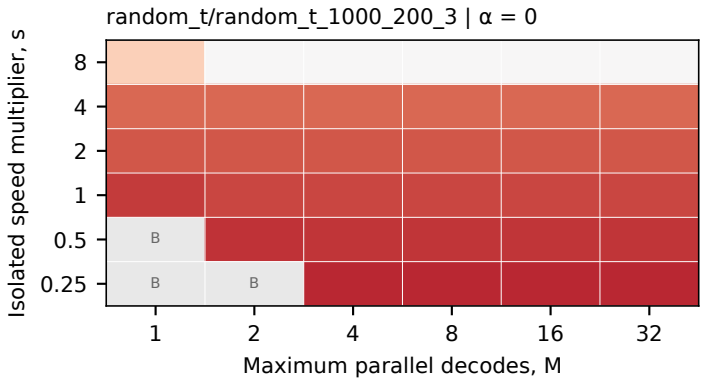


random_t/random_t_1000_200_2 | s = 1

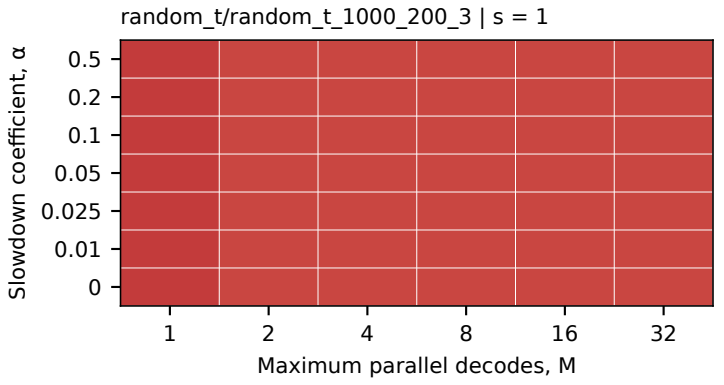


parallel · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

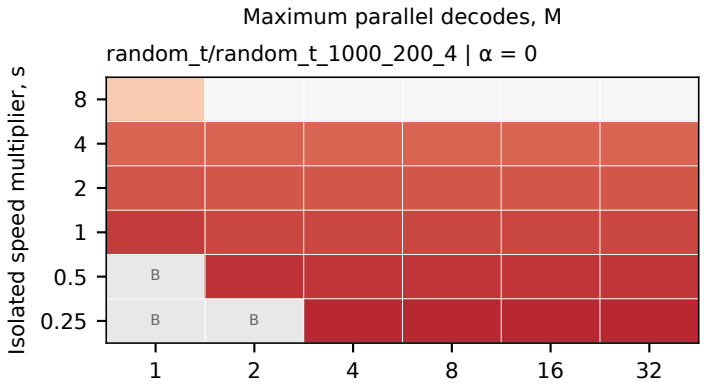
random_t/random_t_1000_200_3 | α = 0



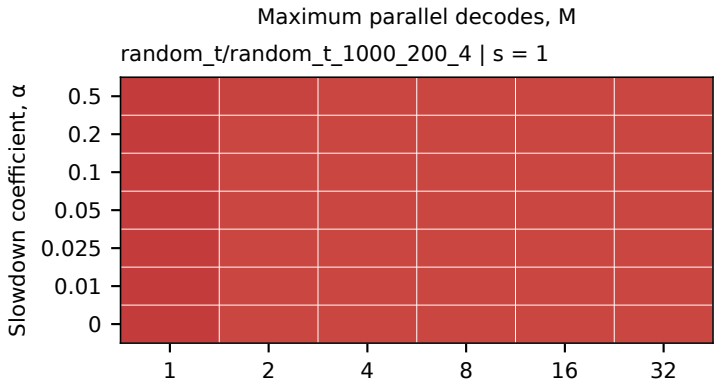
random_t/random_t_1000_200_3 | s = 1



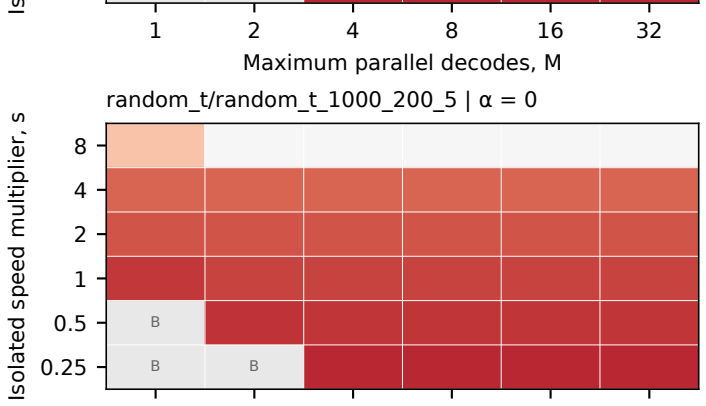
random_t/random_t_1000_200_4 | α = 0



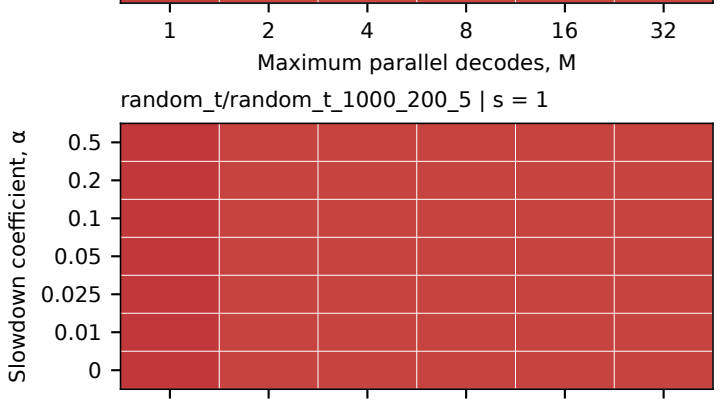
random_t/random_t_1000_200_4 | s = 1



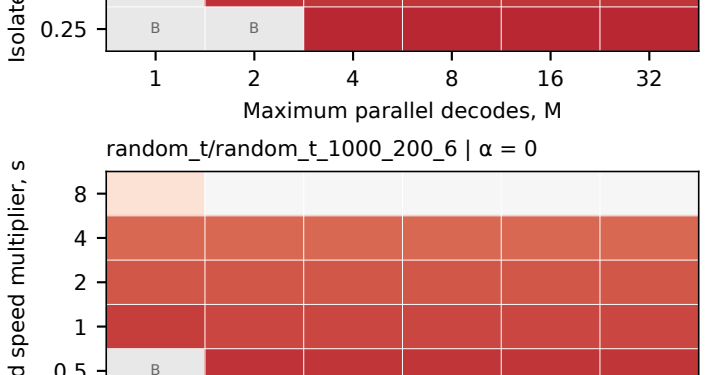
random_t/random_t_1000_200_5 | α = 0



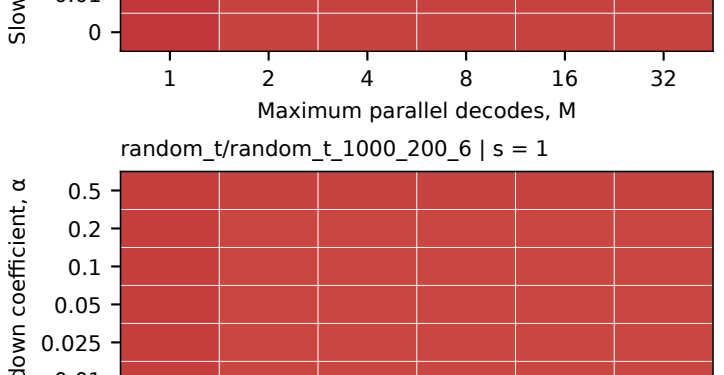
random_t/random_t_1000_200_5 | s = 1



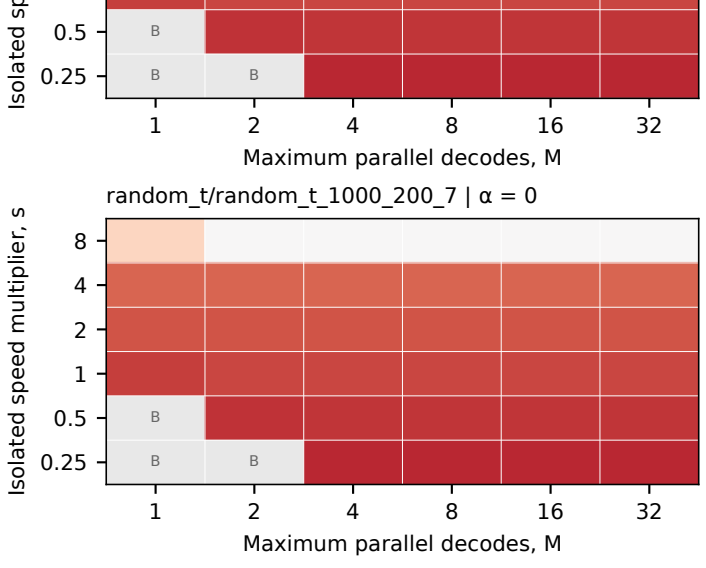
random_t/random_t_1000_200_6 | α = 0



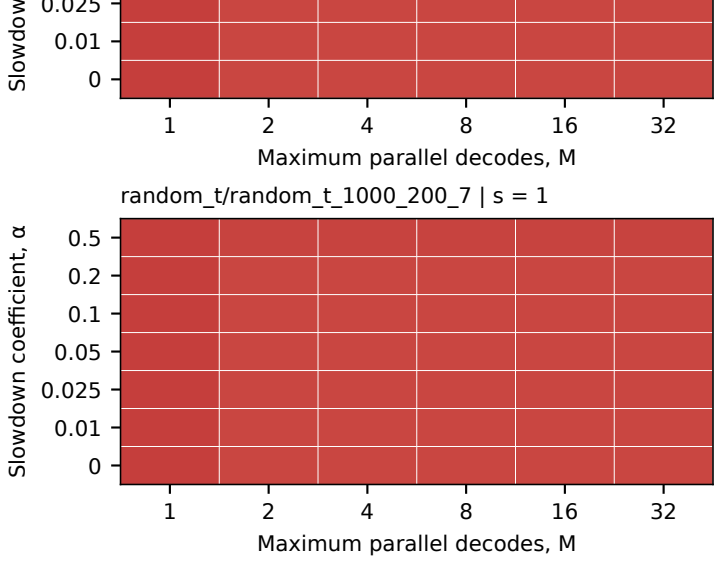
random_t/random_t_1000_200_6 | s = 1



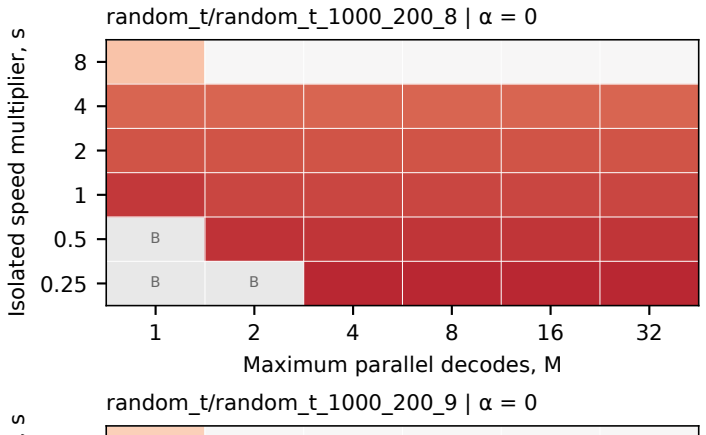
random_t/random_t_1000_200_7 | α = 0



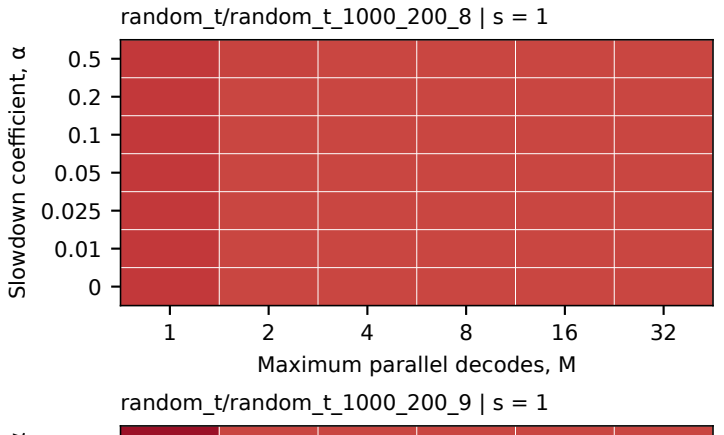
random_t/random_t_1000_200_7 | s = 1



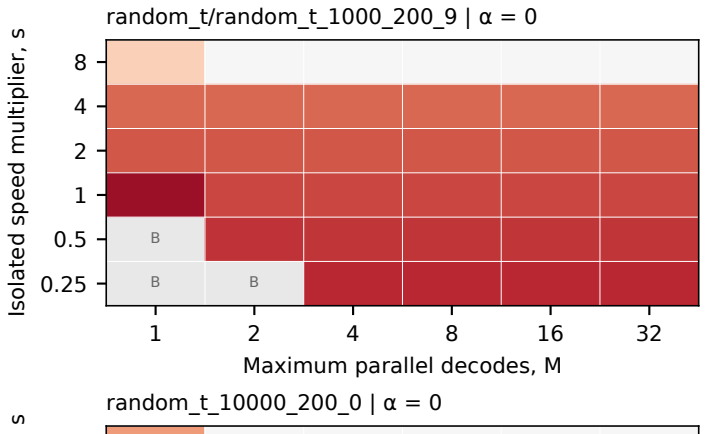
random_t/random_t_1000_200_8 | α = 0



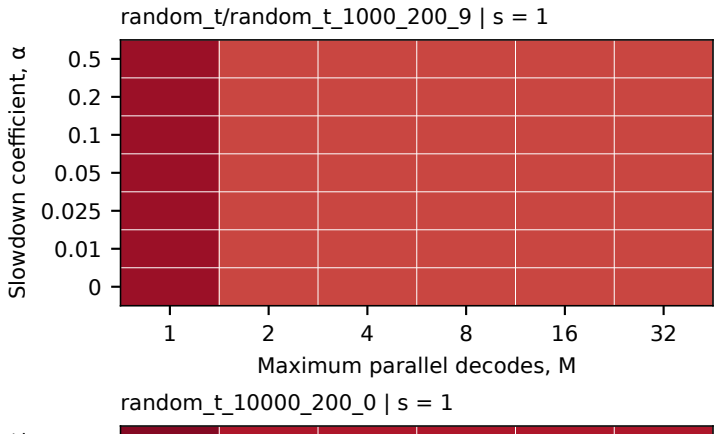
random_t/random_t_1000_200_8 | s = 1



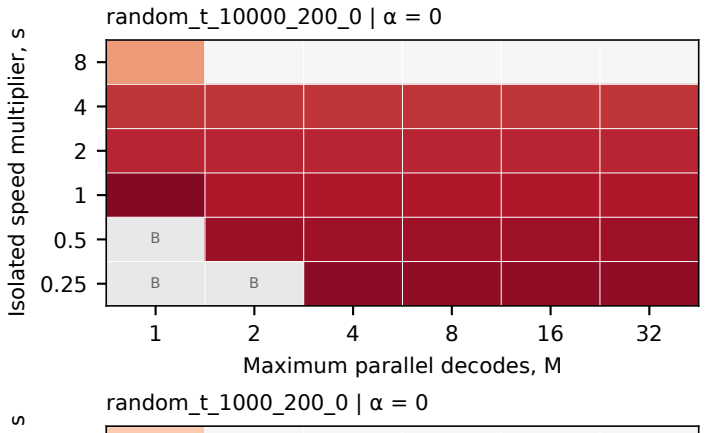
random_t/random_t_1000_200_9 | α = 0



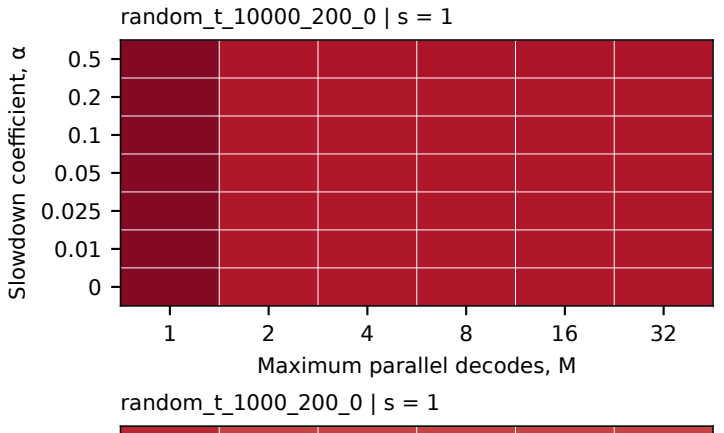
random_t/random_t_1000_200_9 | s = 1



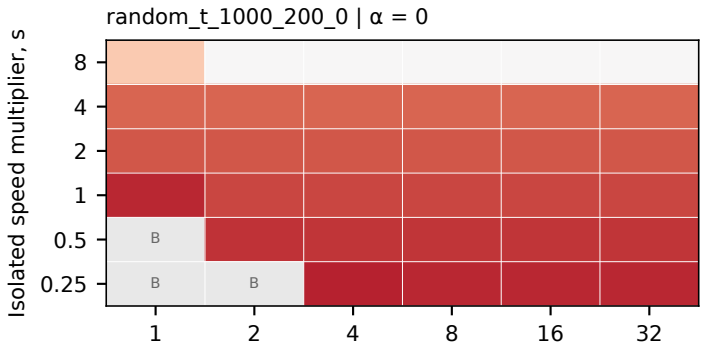
random_t_10000_200_0 | α = 0



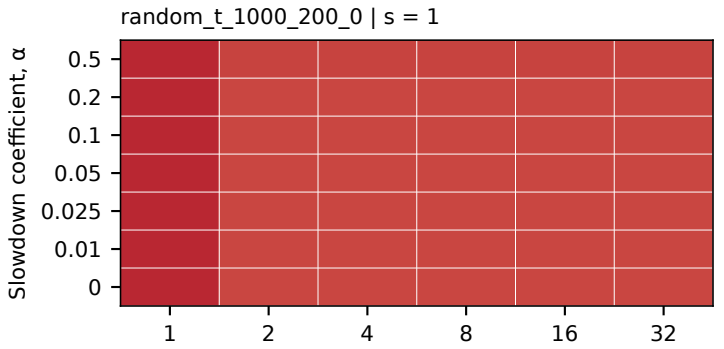
random_t_10000_200_0 | s = 1

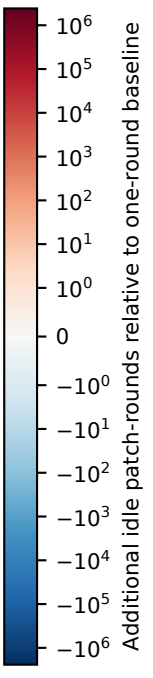


random_t_1000_200_0 | α = 0



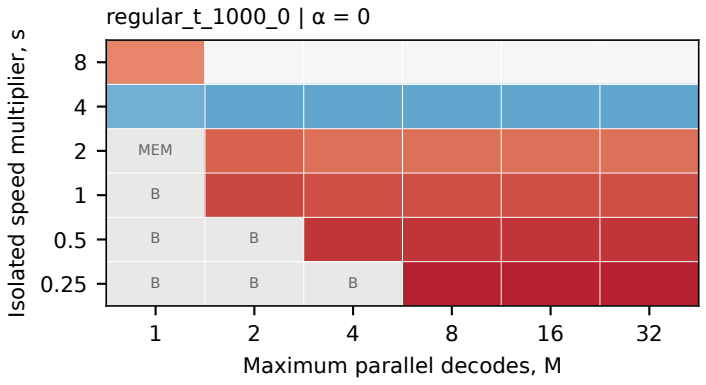
random_t_1000_200_0 | s = 1



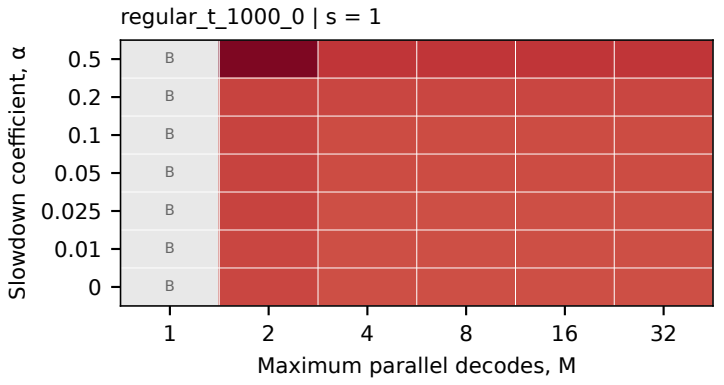


parallel · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

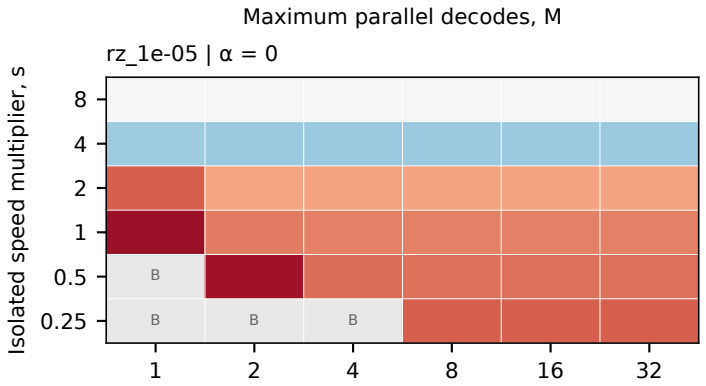
regular_t_1000_0 | α = 0



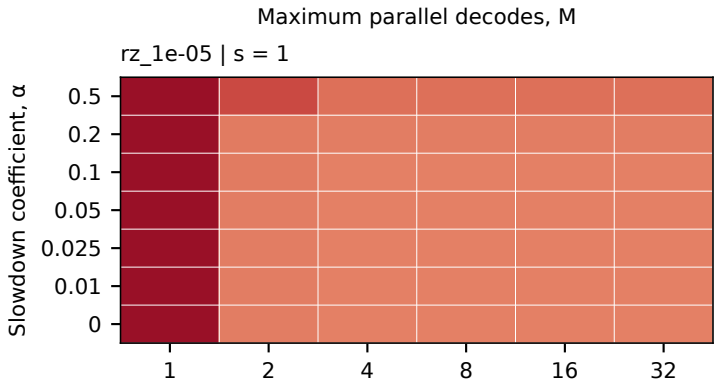
regular_t_1000_0 | s = 1



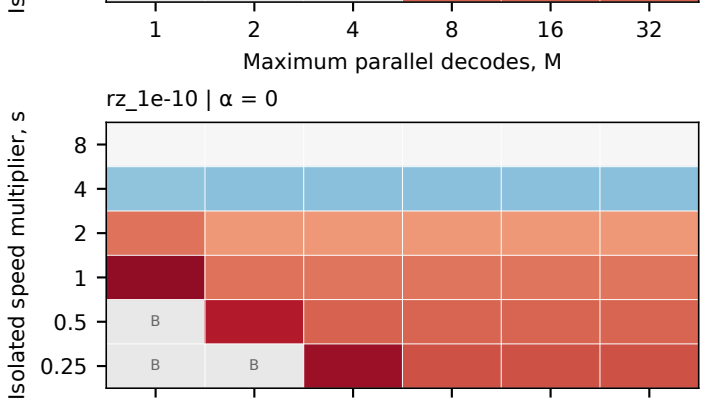
rz_1e-05 | α = 0



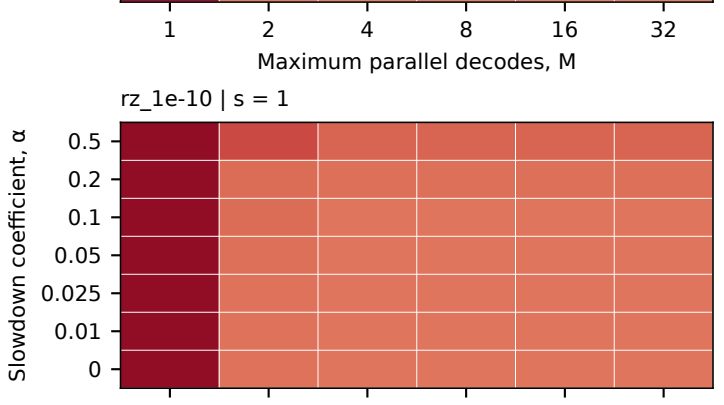
rz_1e-05 | s = 1



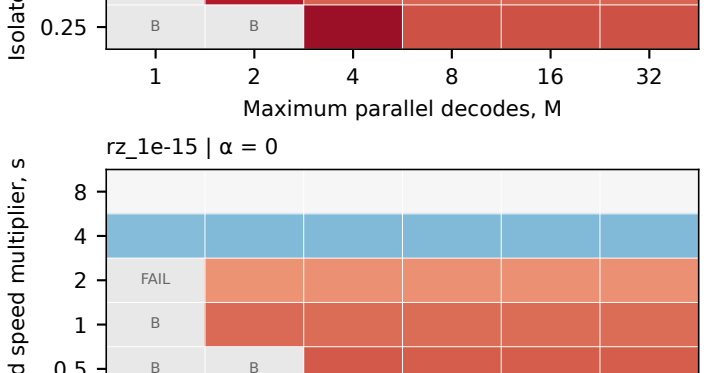
rz_1e-10 | α = 0



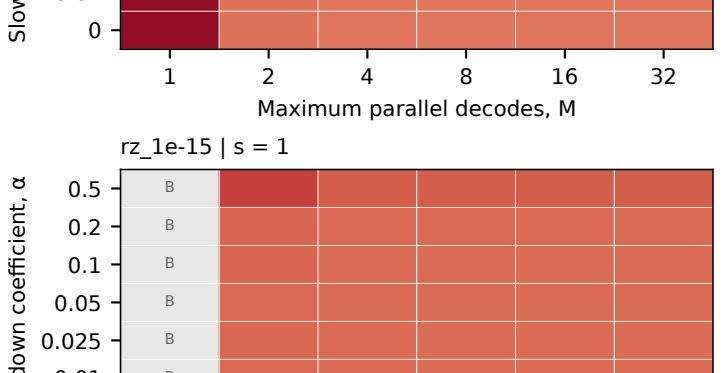
rz_1e-10 | s = 1



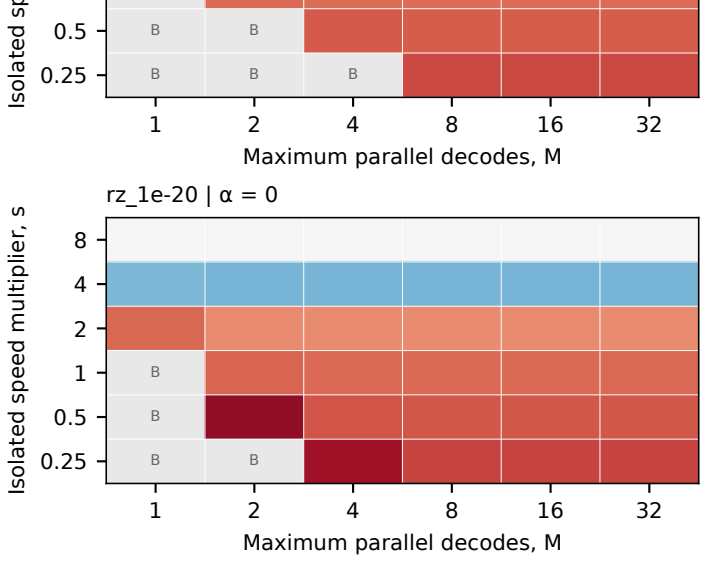
rz_1e-15 | α = 0



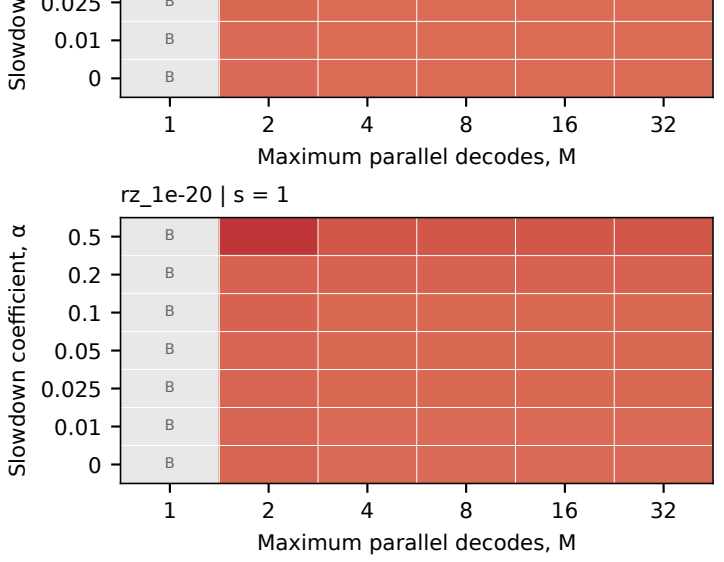
rz_1e-15 | s = 1

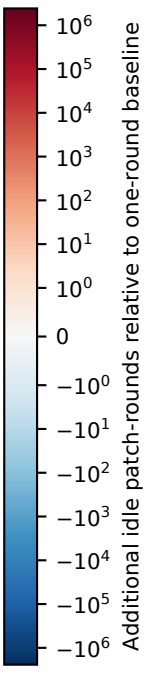


rz_1e-20 | α = 0

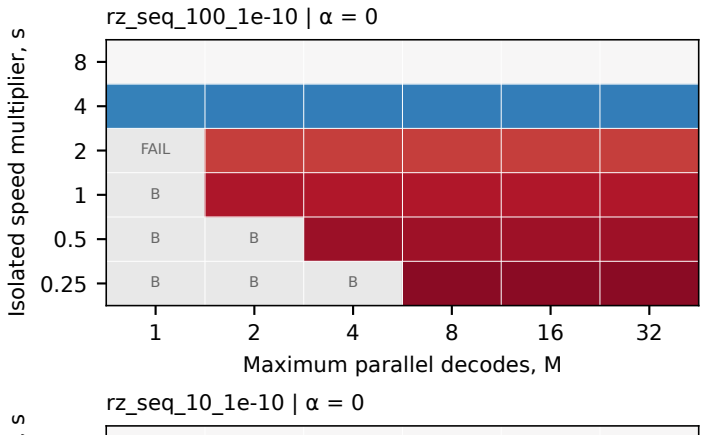


rz_1e-20 | s = 1

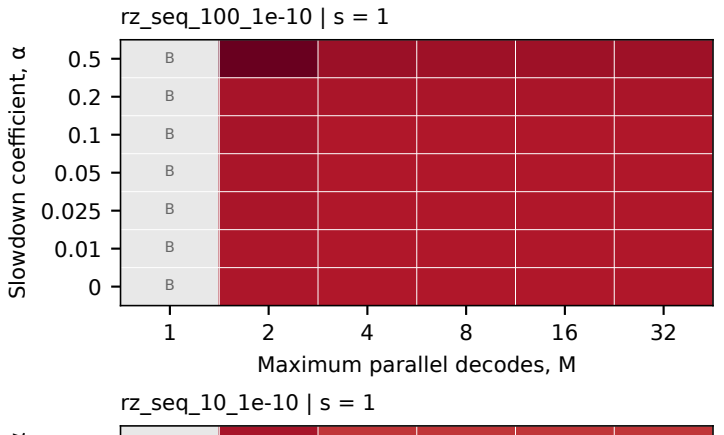




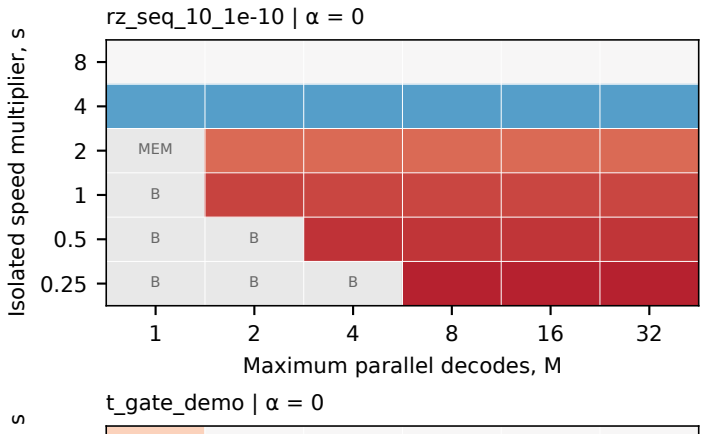
rz_seq_100_1e-10 | α = 0



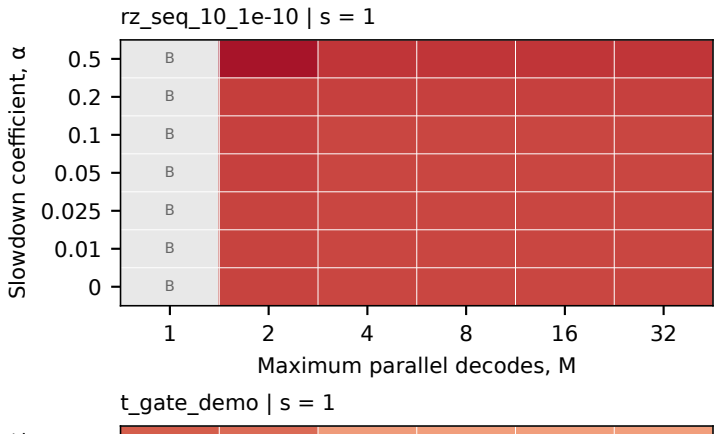
rz_seq_100_1e-10 | s = 1



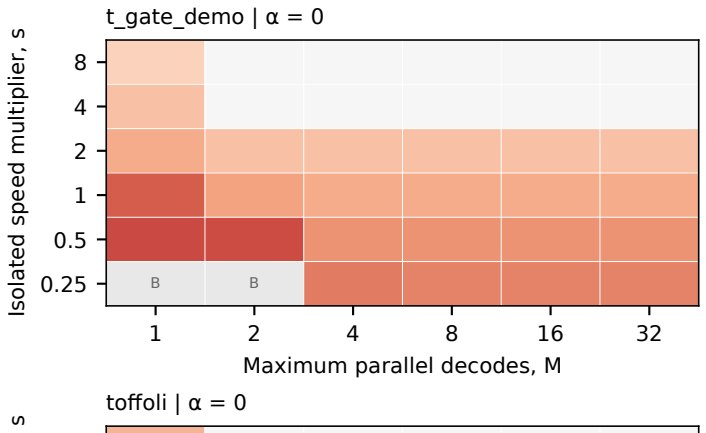
rz_seq_10_1e-10 | α = 0



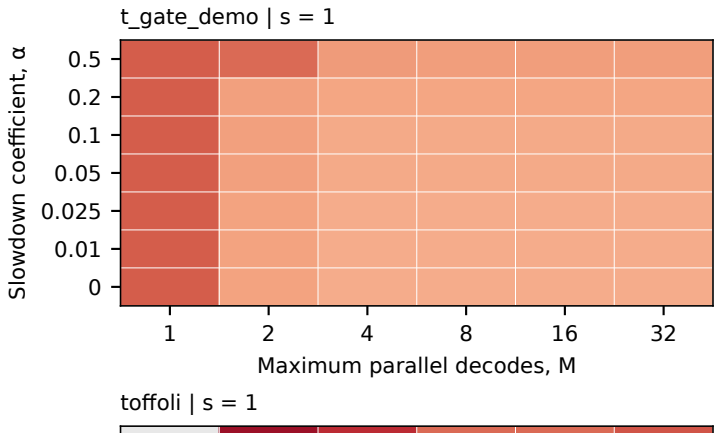
rz_seq_10_1e-10 | s = 1



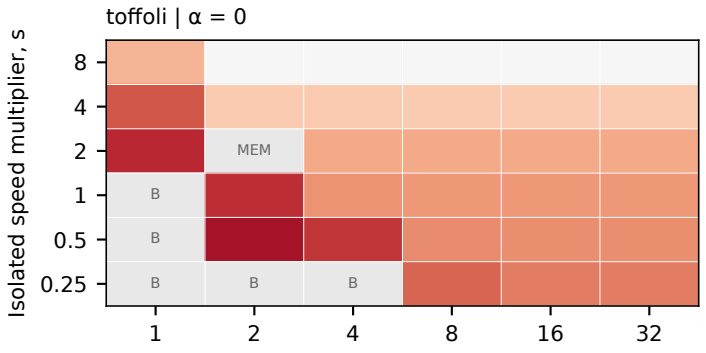
t_gate_demo | α = 0



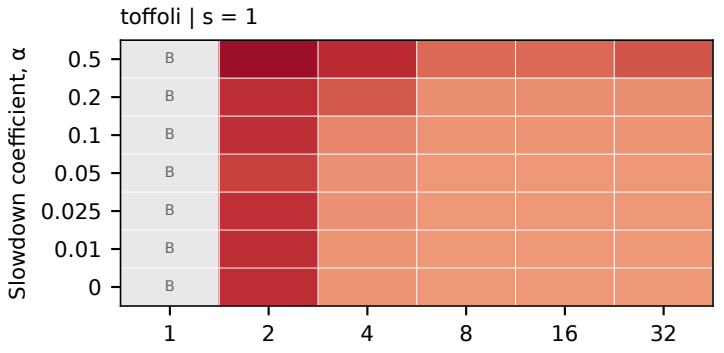
t_gate_demo | s = 1



toffoli | α = 0



toffoli | s = 1



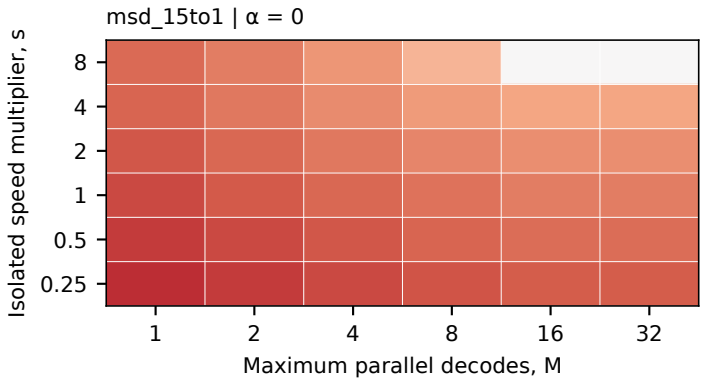
sliding · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

H2 | α = 0
H2 | s = 1
adder_n18 | α = 0
adder_n18 | s = 1
carleman_2_4 | α = 0
carleman_2_4 | s = 1
fermi_hubbard_2_2_Kagome | α = 0
fermi_hubbard_2_2_Kagome | s = 1
fermi_hubbard_2_2_Square | α = 0
fermi_hubbard_2_2_Square | s = 1
fermi_hubbard_4_4_Square | α = 0
fermi_hubbard_4_4_Square | s = 1
grover_ancilla_5 | α = 0
grover_ancilla_5 | s = 1
heisenberg_3 | α = 0
heisenberg_3 | s = 1
memory_10000 | α = 0
memory_10000 | s = 1

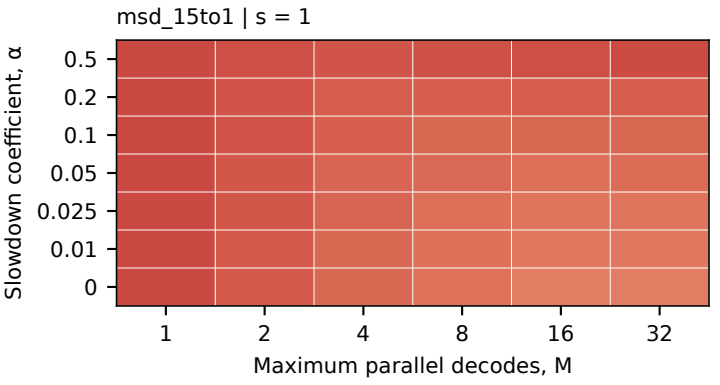
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline

sliding · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

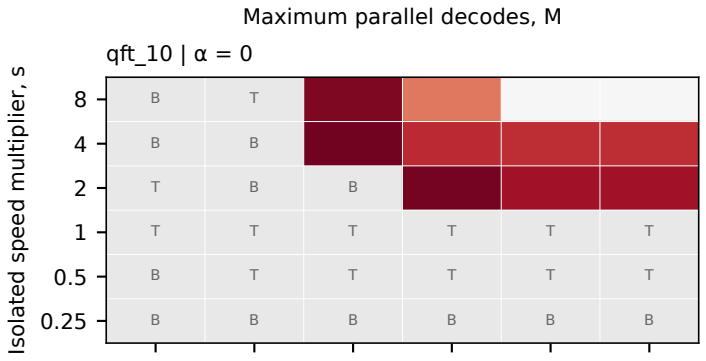
msd_15to1 | α = 0



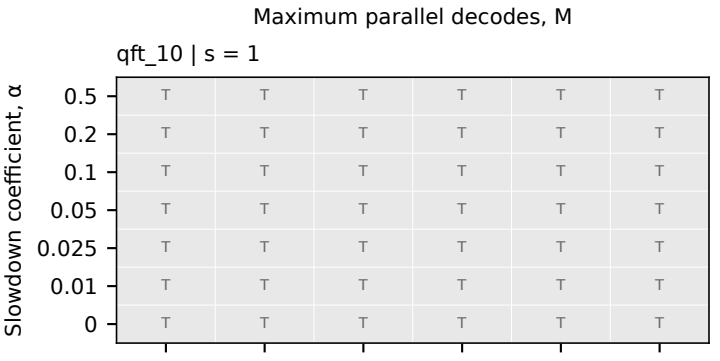
msd_15to1 | s = 1



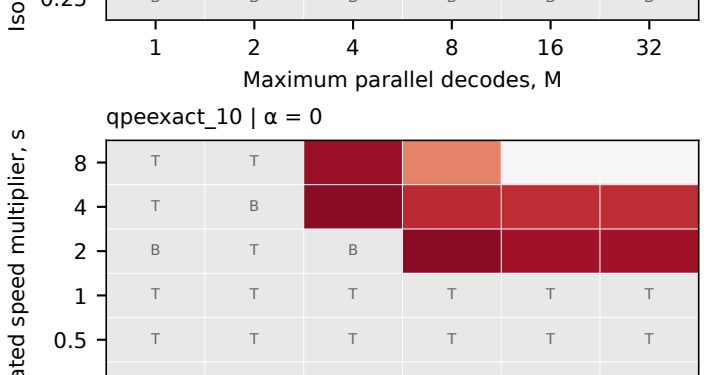
qft_10 | α = 0



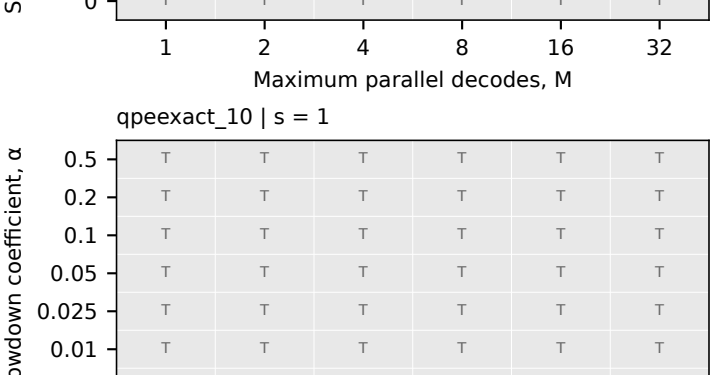
qft_10 | s = 1



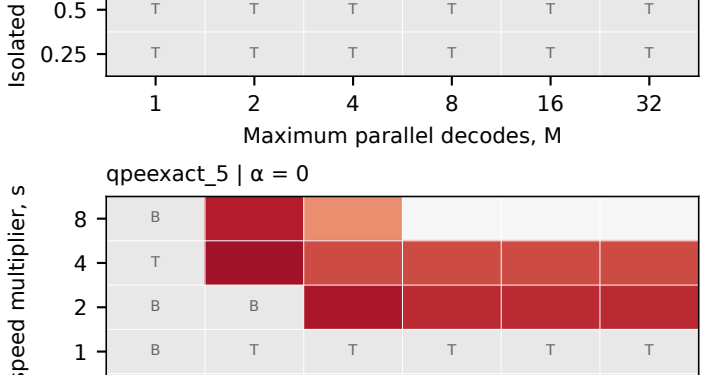
qpeexact_10 | α = 0



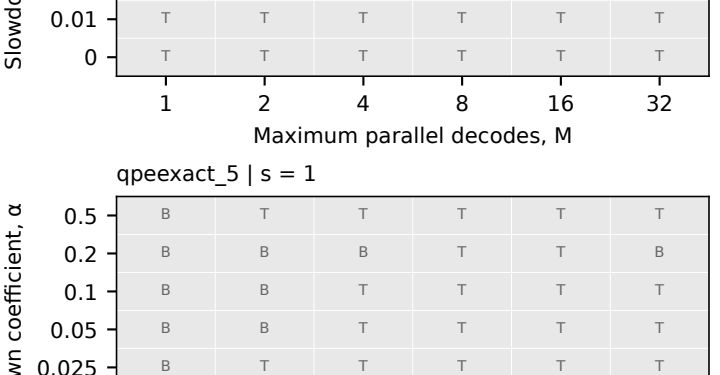
qpeexact_10 | s = 1



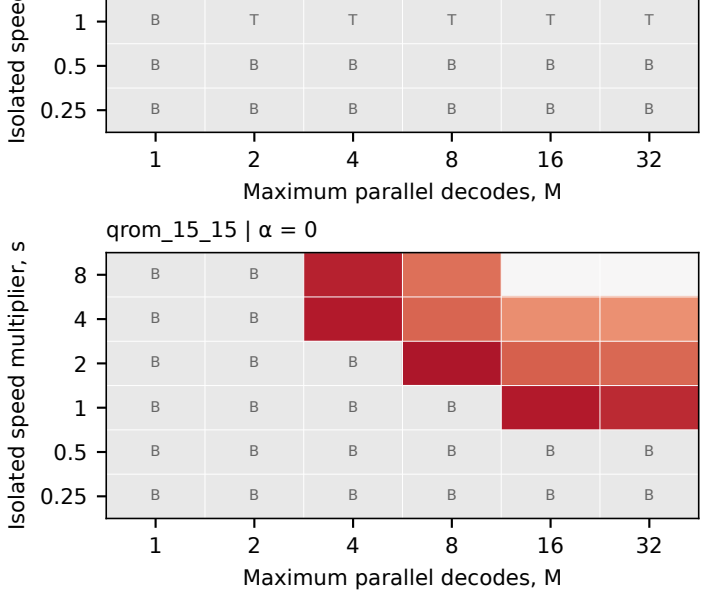
qpeexact_5 | α = 0



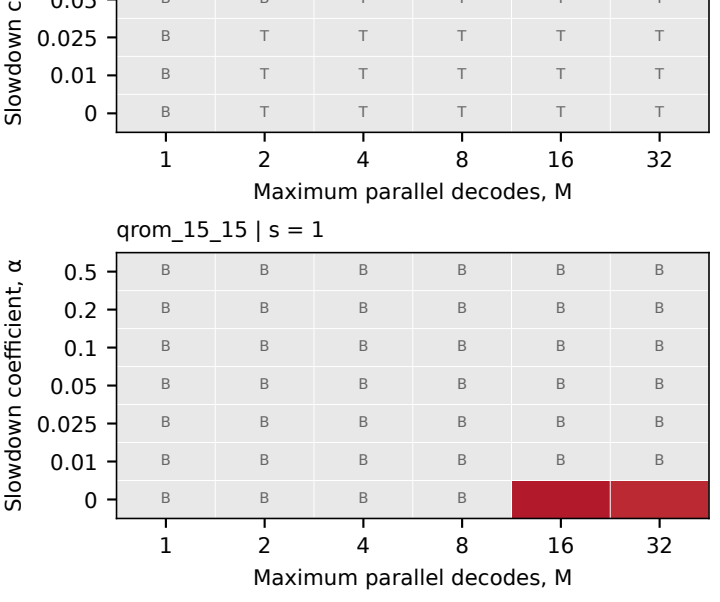
qpeexact_5 | s = 1

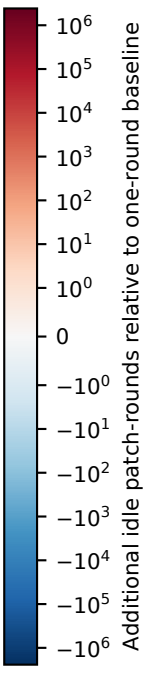


qrom_15_15 | α = 0

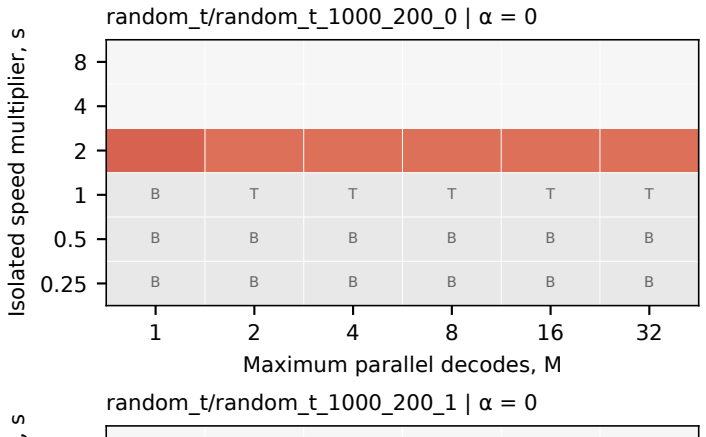


qrom_15_15 | s = 1

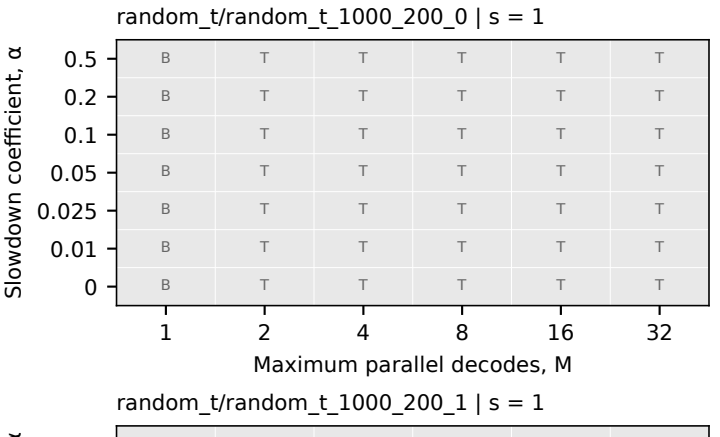




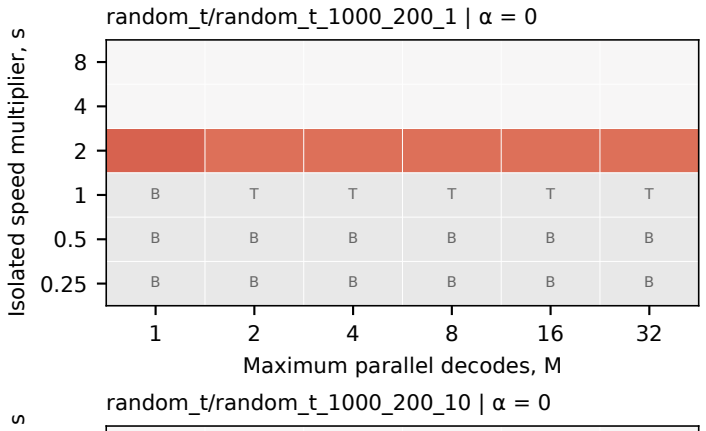
random_t/random_t_1000_200_0 | α = 0



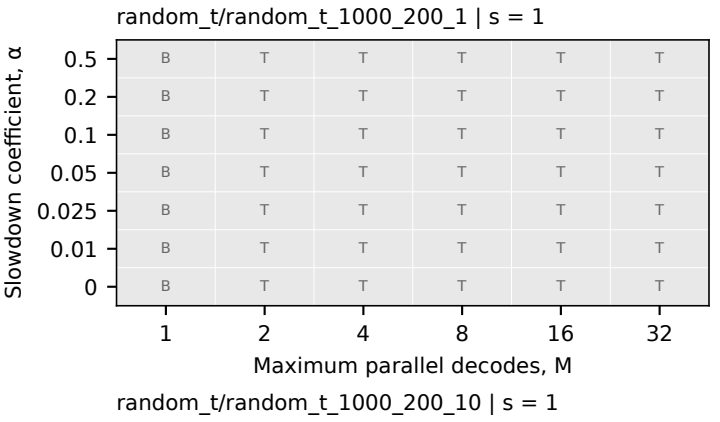
random_t/random_t_1000_200_0 | s = 1



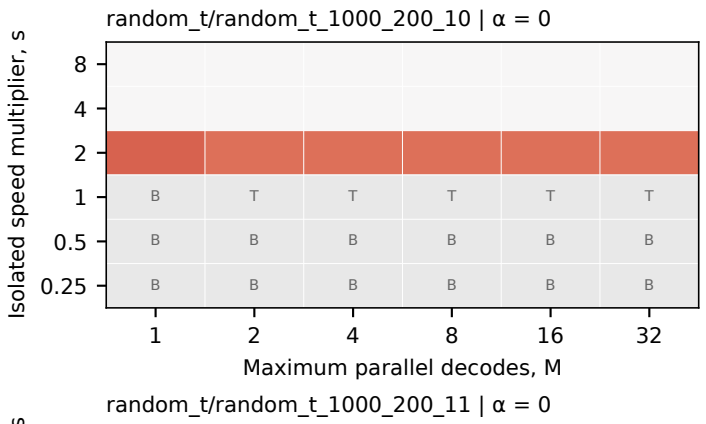
random_t/random_t_1000_200_1 | α = 0



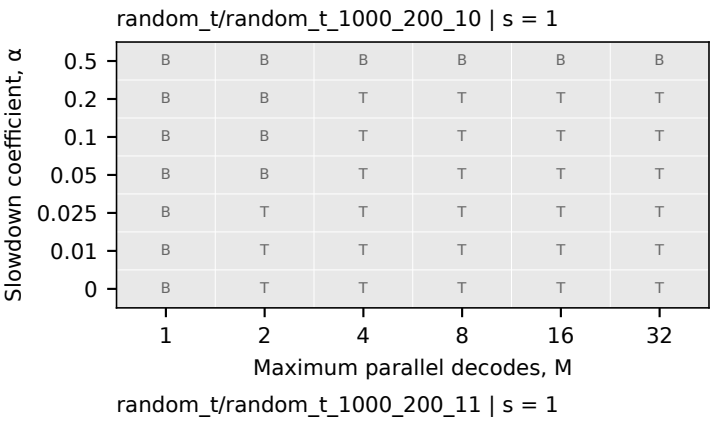
random_t/random_t_1000_200_1 | s = 1



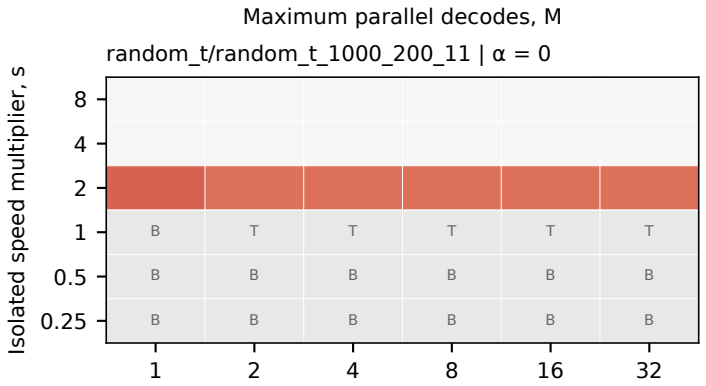
random_t/random_t_1000_200_10 | α = 0



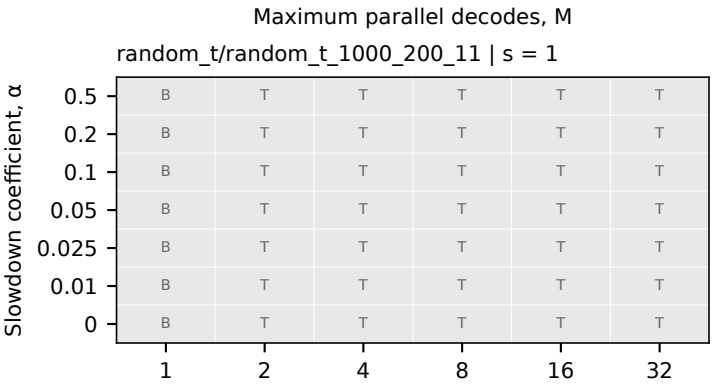
random_t/random_t_1000_200_10 | s = 1



random_t/random_t_1000_200_11 | α = 0



random_t/random_t_1000_200_11 | s = 1



sliding · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

random_t/random_t_1000_200_12 | α = 0
random_t/random_t_1000_200_12 | s = 1
random_t/random_t_1000_200_13 | α = 0
random_t/random_t_1000_200_13 | s = 1
random_t/random_t_1000_200_14 | α = 0
random_t/random_t_1000_200_14 | s = 1
random_t/random_t_1000_200_15 | α = 0
random_t/random_t_1000_200_15 | s = 1
random_t/random_t_1000_200_16 | α = 0
random_t/random_t_1000_200_16 | s = 1
random_t/random_t_1000_200_17 | α = 0
random_t/random_t_1000_200_17 | s = 1
random_t/random_t_1000_200_18 | α = 0
random_t/random_t_1000_200_18 | s = 1
random_t/random_t_1000_200_19 | α = 0
random_t/random_t_1000_200_19 | s = 1
random_t/random_t_1000_200_2 | α = 0
random_t/random_t_1000_200_2 | s = 1

Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline

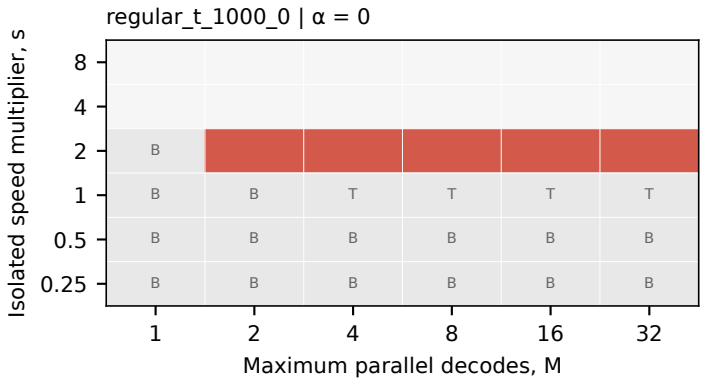
sliding · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

random_t/random_t_1000_200_3 | α = 0
random_t/random_t_1000_200_3 | s = 1
random_t/random_t_1000_200_4 | α = 0
random_t/random_t_1000_200_4 | s = 1
random_t/random_t_1000_200_5 | α = 0
random_t/random_t_1000_200_5 | s = 1
random_t/random_t_1000_200_6 | α = 0
random_t/random_t_1000_200_6 | s = 1
random_t/random_t_1000_200_7 | α = 0
random_t/random_t_1000_200_7 | s = 1
random_t/random_t_1000_200_8 | α = 0
random_t/random_t_1000_200_8 | s = 1
random_t/random_t_1000_200_9 | α = 0
random_t/random_t_1000_200_9 | s = 1
random_t_10000_200_0 | α = 0
random_t_10000_200_0 | s = 1
random_t_1000_200_0 | α = 0
random_t_1000_200_0 | s = 1

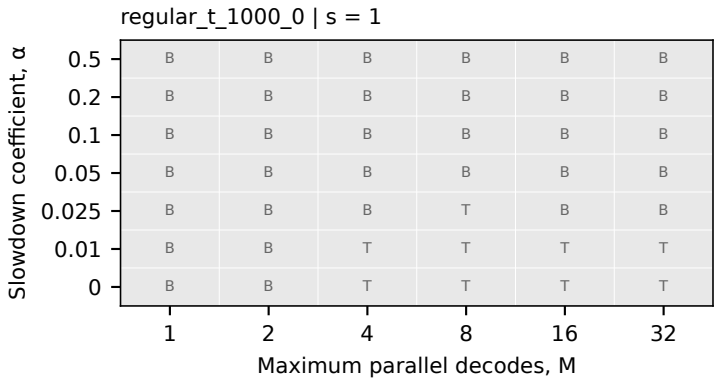
Isolated speed multiplier, s
Slowdown coefficient, α
Maximum parallel decodes, M
Additional idle patch-rounds relative to one-round baseline

sliding · shared scale across all benchmarks
CPU PyMatching surrogate · d = 3 · QEC round = 1 µs · speculation off

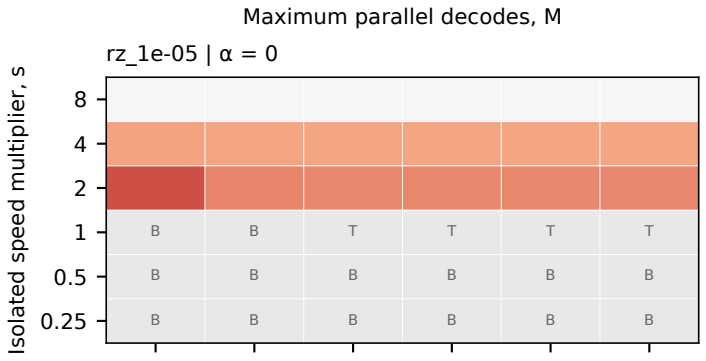
regular_t_1000_0 | α = 0



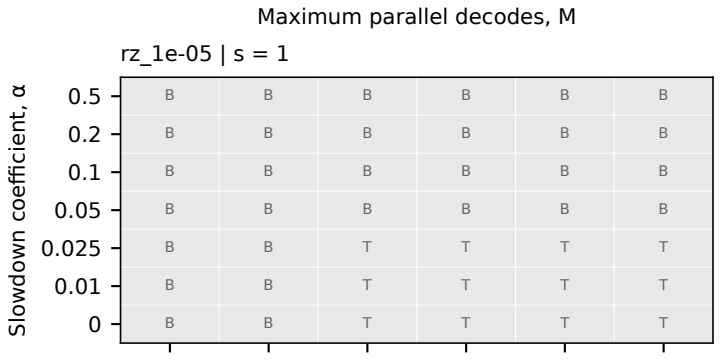
regular_t_1000_0 | s = 1



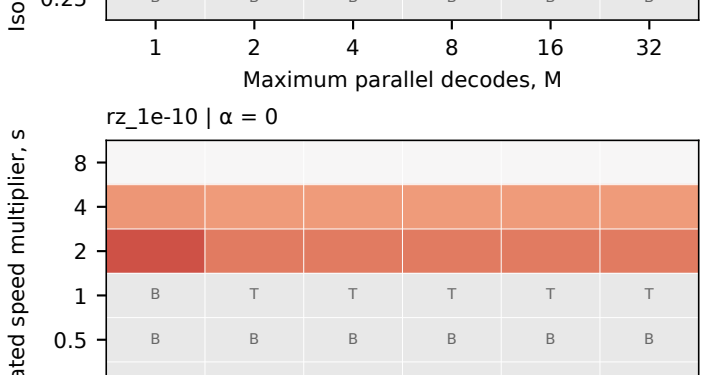
rz_1e-05 | α = 0



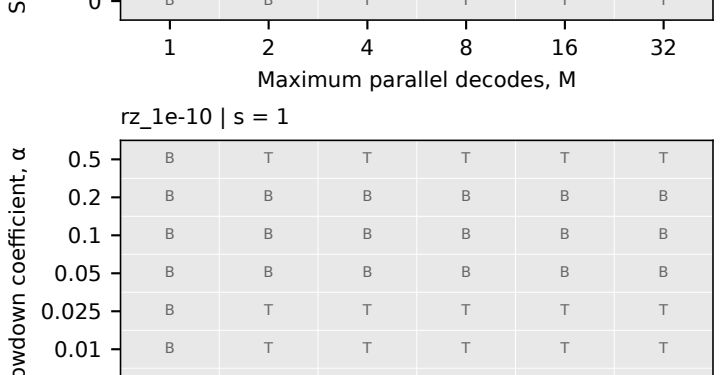
rz_1e-05 | s = 1



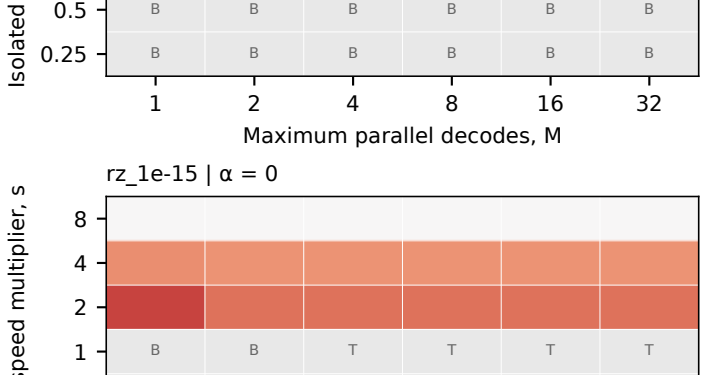
rz_1e-10 | α = 0



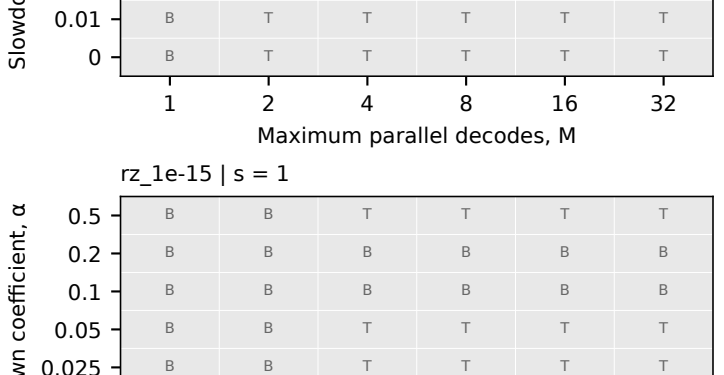
rz_1e-10 | s = 1



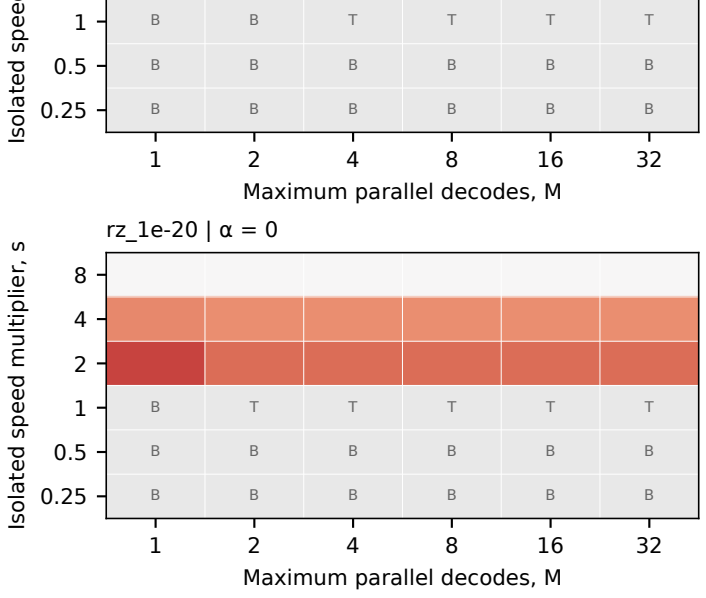
rz_1e-15 | α = 0



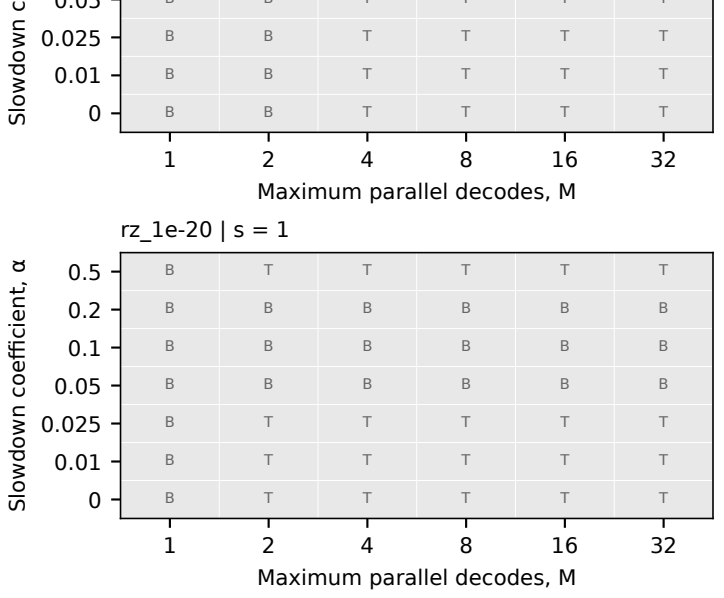
rz_1e-15 | s = 1

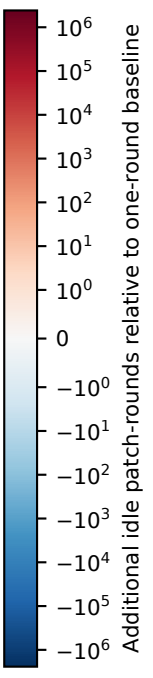


rz_1e-20 | α = 0

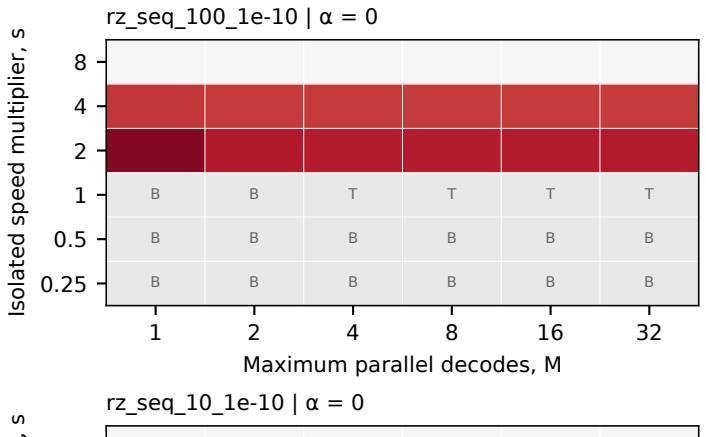


rz_1e-20 | s = 1

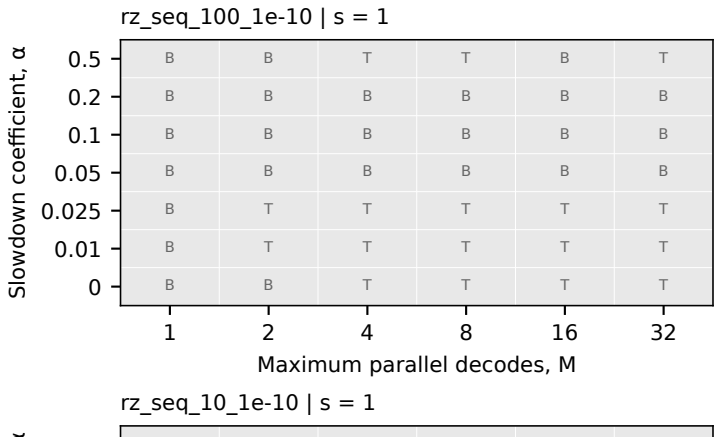




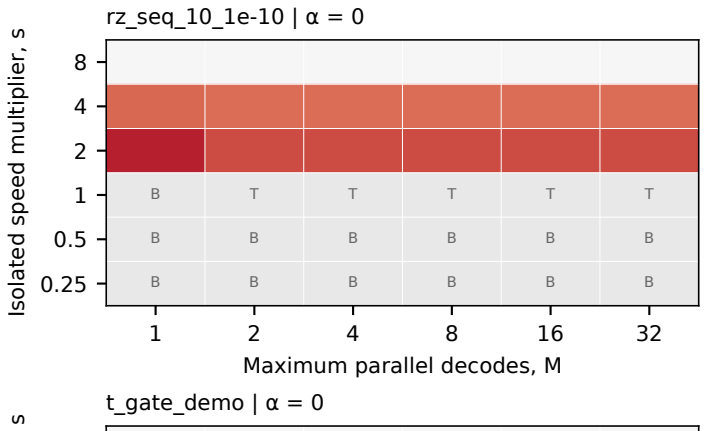
rz_seq_100_1e-10 | α = 0



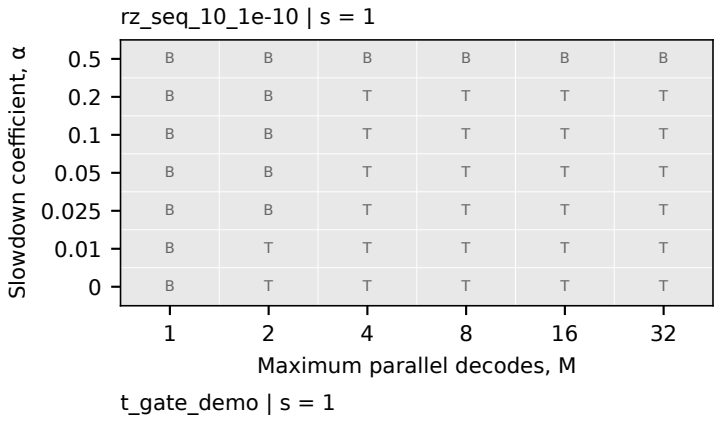
rz_seq_100_1e-10 | s = 1



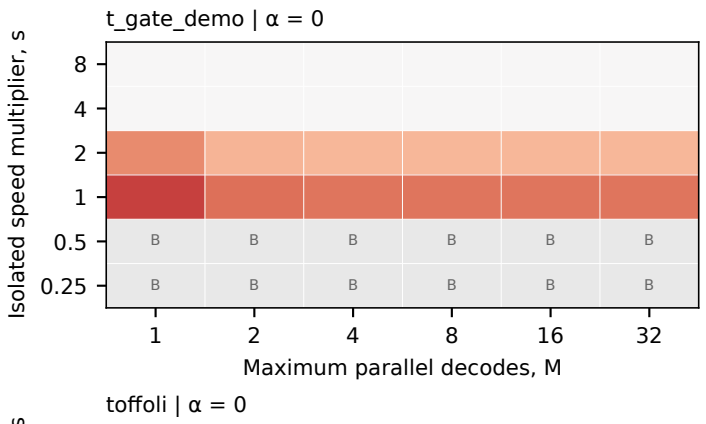
rz_seq_10_1e-10 | α = 0



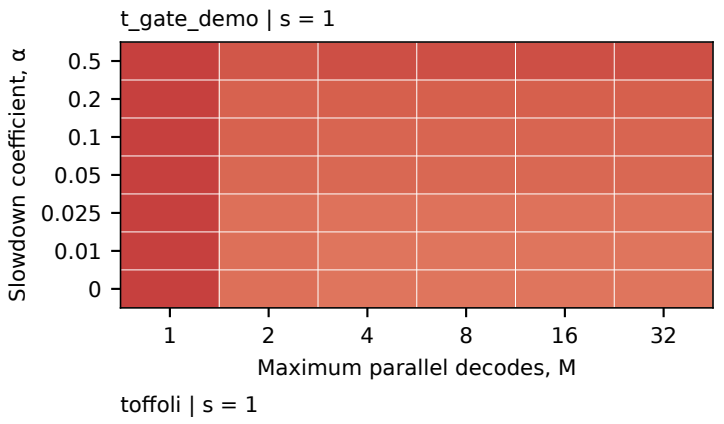
rz_seq_10_1e-10 | s = 1



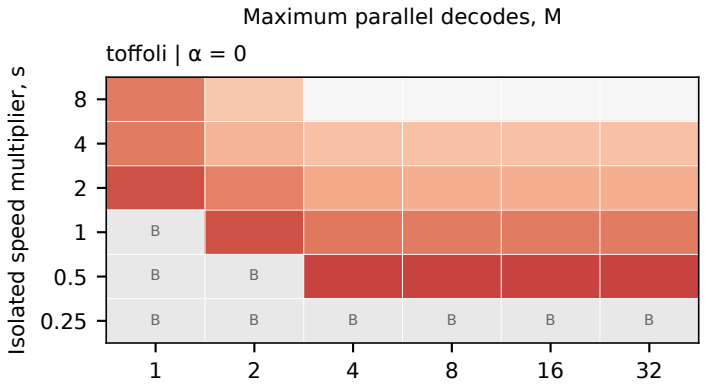
t_gate_demo | α = 0



t_gate_demo | s = 1



toffoli | α = 0



toffoli | s = 1

